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Master in Business Analytics & Data Science
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The Price of Sophistication: A Pre-Committed Falsification Test for When Spatial and Robust Modelling Pay in Carbon-Aware Data-Center Scheduling

A Day-Ahead Migration Scheduler and a Complexity–Value Decision Rule

Marco Ortiz Togashi

Supervisor: Prof. Bissan Ghaddar

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AI Use Declaration. Generative AI tools were used as a research and engineering assistant under the author's direction: for code implementation support, experiment orchestration, literature triage, and drafting assistance. All research questions, modelling decisions, experimental designs, and interpretations are the author's, developed with the supervisor; all results were produced by version-controlled code reviewed by the author (with the Phase 1–2 falsification protocol pre-committed; the Part 3 transfer and emergency-crossover sweeps are exploratory extensions, not part of that pre-commitment), and all text was reviewed and edited by the author.

Ethics Statement. This research involves no human subjects, personal data, or animal experimentation, and required no ethics-board approval. It uses only grid-level carbon-intensity estimates (Electricity Maps, under an academic licence that prohibits redistribution of the raw series, respected here) and public reanalysis weather data (Open-Meteo, CC-BY). The work is motivated by an environmental aim, reducing the carbon footprint of flexible compute, and the author reports the negative and adverse results (the dependence-modelling null and the cases where robustness loses) as candidly as the positive ones, to avoid overstating the value of the proposed methods. The author declares no conflicts of interest.

Abstract

Data-center operators can shift compute toward cleaner hours and regions, and recent work applies distributionally robust optimization (DRO) to carbon-aware scheduling. The unexamined question is which modelling layers actually pay for their complexity. This thesis builds a working day-ahead carbon-aware migration scheduler and characterises the value of each layer. Over the honest baseline (carbon-aware scheduling with no inter-region transfer, $\Phi = 0$, on the same feasible set), active migration cuts daily $\mathrm{CVaR}_{0.95}$ emissions by 4.0–9.9% (Western 4.0%, Eastern 9.9%, Diversified 9.0%), and this deterministic transfer channel is the dominant lever; the carbon-computing literature independently finds spatial shifting dominates temporal shifting. This thesis introduces a pre-committed, falsification-based methodology for *pricing* each modelling layer, separating the *validity* of the spatial-dependence assumption (the cross-region carbon correlations are real) from its *value* to the decision (a replicated null). We then isolate the value of *passive* sophistication. Using a Mahalanobis–Wasserstein DRO solved as a second-order cone program, a pre-committed shuffled-marginals falsification test across three real US/Canada grids spanning the dependence spectrum (the US West, an Ontario-anchored Eastern belt, and an engineered solar/wind/hydro portfolio) returns a replicated **null**. The encoded cross-region covariance adds below 0.4% of $\mathrm{CVaR}_{0.95}$, and the null survives Ledoit–Wolf shrinkage, removing each region's average daily pattern (residualization), Benjamini–Hochberg correction, walk-forward validation, and a per-cell equivalence test. A mean-leveling test shows the covariance signal is genuine but masked by the mean field (a mean-dominance bound), and a tail-dependence analysis shows the residual dependence is non-elliptical and upper-tail independent, hence invisible to covariance-based ambiguity sets; a one-paragraph copula appendix confirms even the maximal coupling recovers nothing. Finally we ask when robustness itself pays. Following the price of robustness of Bertsimas and Sim, a DRO over day-ahead forecast error buys a small worst-day ($\mathrm{CVaR}_{0.95}$) tail reduction at a mean premium, with a value of the stochastic solution near zero, and pays over deterministic transfer only past an emergency-severity crossover $M^{\star} \approx 3$ (with M the multiple by which a region's carbon intensity spikes in a rare grid emergency, so $M = 3$ is a tripling); tested against real data-grounded worst-tail emergencies across 17 zones, that crossover does not activate. The thesis contributes a working scheduler with honest savings, a screening rule for when passive dependence modelling fails, and a price-of-robustness decision rule with a tested, data-grounded bound, all under 202 unit tests, with the Phase 1–2 falsification protocol pre-committed (the stylised-emergency crossover of RQ3 is a calibrated stress test, not part of the pre-commitment).

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Nomenclature

Symbols used throughout; plain-language glosses are in Appendix D.

Symbol	Meaning
$R,\ T$	number of regions; hours in the day-ahead horizon ($T = 24$)
$x \in \mathbb{R}^{R\times T}$	schedule (compute allocated per region and hour) [MW]
$y_{r,t}$	executed load after inter-region migration [MW]
W_r	day-ahead workload to be served by region r [MWh]
$\rho \in \mathbb{R}^{R\times T}$	uncertain carbon-intensity field [gCO_2eq/kWh]
$\bar{\rho}$	mean (forecast) carbon field; $\rho = \bar{\rho} + \xi$
ξ	zero-mean carbon residual (the dependence-bearing part)
Σ	cross-region space–time covariance of ρ ($RT \times RT$)
L	Cholesky factor, $LL^\top = \Sigma$
ε	Wasserstein ambiguity radius (DRO ball size)
η	epigraph (cone) variable in the second-order cone program
λ	covariance weight in the mean-variance stress objective (Sec. 4.8)
α_r	inflexible fraction of region r's workload
Φ	inter-region transfer budget ($\Phi = 0$ is the no-transfer baseline)
$f_{r\to s,t}$	migrated load from region r to s at hour t [MW]
CVaR_β	Conditional Value-at-Risk at level β ($\beta = 0.95$)
τ	Value-at-Risk threshold in the CVaR program
$M,\ M^\star$	emergency severity multiple; the crossover severity
$\chi_{\mathrm{L}},\ \chi_{\mathrm{U}}$	lower / upper tail-dependence coefficients
Λ	dependence leverage (comonotone minus diversified tail spread)
$\mathrm{OPT}(\Sigma)$	optimal robust value of the SOCP under covariance Σ

1 Introduction

1.1 Context

Data centers consumed an estimated 1–2% of global electricity, and demand is growing rapidly with AI workloads [10]. Unlike most industrial loads, compute is unusually *flexible*: many batch and training workloads can be delayed by hours or moved between sites with little loss of utility. This movement is *logical*. Jobs relocate over the network between an operator's data centers and do not require the regions' power grids to be physically interconnected, so cross-region correlation of carbon intensity matters as information for the scheduler, not as an electrical constraint.[1] The carbon intensity of grid electricity varies strongly by hour and region with wind, solar, hydro availability, and demand, and this flexibility lets an operator act as a grid-aware participant that defers work from dirty hours to clean ones [20, 23]. Google's carbon-intelligent computing platform demonstrated such temporal shifting at production scale [20], and recent systems work extends it to variable-capacity operation [24].

Scheduling against *tomorrow's* carbon intensity, however, means scheduling under uncertainty. Distributionally robust optimization (DRO) addresses this by optimizing against the worst distribution in an ambiguity ball around the empirical distribution [8], and has been applied to carbon-aware scheduling with carbon intensity as the uncertain quantity [9]. Existing formulations treat each region's carbon intensity essentially in isolation, yet the carbon intensities of electrically interconnected regions co-move: weather fronts traverse neighbouring balancing areas, and shared solar fleets create synchronized clean periods. We therefore model carbon intensity as a stochastic *vector* with a full cross-region covariance.

1.2 Problem statement and research questions

Each modelling layer (cross-region covariance, distributional robustness, richer copulas) adds estimation and computational cost and is worthwhile only if it buys measurably better schedules. The literature has assumed, rather than measured, that this sophistication pays. This thesis builds the scheduler and measures the value of each layer directly.

RQ1. How much can a day-ahead carbon-aware migration scheduler cut daily $\mathrm{CVaR}_{0.95}$ emissions relative to carbon-aware scheduling without inter-region transfer, and which lever drives the saving?

[1] We use "spatial" throughout in the sense of *across the operator's geographically separate regions* (sites), as in the standard spatial-versus-temporal load-shifting distinction. The dependence we study is the cross-region statistical correlation of carbon intensity, driven by shared weather and generation; it is not distance-decay spatial autocorrelation (e.g. Moran's I or a variogram) and not an electrical coupling of the grids, which need not be interconnected.

RQ2. When does adding passive modelling complexity (cross-region covariance, or a non-elliptical copula) improve out-of-sample tail emissions, and at what cost?

RQ3. When does robustifying against day-ahead forecast error pay over deterministic transfer, and do real grids ever reach that regime? What decision rule follows?

The value question for RQ2 was specified and commit-locked in version control as a falsifiable hypothesis test before the held-out 2025 year was evaluated (the "pre-committed" design; Section 3.7):

Pre-committed hypotheses (Phase 1–2, RQ2).

H_1 (modelling pays): encoding cross-region carbon dependence in the ambiguity set lowers out-of-sample $\mathrm{CVaR}_{0.95}$ by more than a 0.4% materiality margin (the smallest tail saving an operator would act on), relative to the dependence-destroyed (shuffled-marginals) model on the same feasible set.

H_0 (the spatial null): it does not; the out-of-sample spatial gap lies within ±0.4%.

Decision rule: because one cannot simply accept a null, we test it by equivalence. The formal equivalence-test (TOST) null is that a material effect *is* present ($|\text{gap}| \geq 0.4\%$); affirmatively rejecting it in every pre-committed grid and constraint regime is what licenses the spatial null H_0. A mean-leveling positive control must succeed (detect value where value is constructed to exist) for the null to count as a true negative rather than an underpowered test.

1.3 Contributions

The thesis makes five contributions: one pre-committed and confirmatory (the dependence null, RQ2), one a baseline-anchored measurement (the day-ahead savings, RQ1), one methodological, one exploratory (the robustness crossover, RQ3), and one infrastructural. (i) *A working day-ahead scheduler with honest, baseline-anchored savings*: active inter-region migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% over the no-transfer ($\Phi = 0$) baseline on the same feasible set across three real grids (a gross spatial ceiling, reported before migration cost). The transfer mechanism follows established carbon-aware practice; the contribution is the $\Phi = 0$-anchored measurement of its spatial value. (ii) *A pre-committed methodology for pricing modelling layers*: holding the optimization fixed and manipulating one object at a time, it measures the marginal out-of-sample CVaR value of each layer, and separates the *validity* of the spatial-dependence assumption (correlations are genuine, 0.7–0.8) from its *value* to the decision (null). To our knowledge this is the first treatment to place carbon intensity itself in a Mahalanobis–Wasserstein ball as a correlated cross-region vector and isolate the marginal value of that coupling via a shuffled-marginals falsification test. (iii) *A screening rule* for when passive dependence modelling cannot help, supported from four independent directions: an a-priori SOCP gap-bound (Proposition 1) before any data is touched; a mean-leveling positive control showing the covariance signal is genuine but masked by the mean field; a tail-dependence/comonotone-copula argument showing even the maximal coupling recovers nothing; and a risk-measure swap showing the null survives replacing the standard-deviation penalty with a variance penalty built to favour covariance (Section 4.8). (iv) *A price-of-robustness decision rule*, reported as an *exploratory* extension outside the pre-commitment, with a data-grounded, falsified crossover: robustification pays only past an emergency-severity crossover $M^{\star} \approx 3$ that realized severity (1.3–1.9×; Winter Storm Uri 1.3×) does not reach on observed data, stated as the rule plus its tested bound, not a universal impossibility. (v) *A reproducible, validated pipeline*: pre-commitment, 202 unit tests, a small-instance model validation, a robustness battery, Benjamini–Hochberg correction, and per-cell TOST equivalence tests that affirmatively reject any effect above the 0.4% materiality margin.

1.4 Thesis structure

Section 2 reviews the carbon-aware scheduling and DRO literature and locates the gap: the value of modelling sophistication has been assumed, not measured. Section 3 develops the day-ahead migration scheduler, the DRO model, the falsification test, and the evaluation protocol. Section 4 reports the scheduler savings and the value of each modelling layer, the screening rule for passive dependence, and the price-of-robustness decision rule with the severity crossover. Section 5 interprets the findings and states limitations, and Section 6 concludes with the practitioner decision rule and future work. Appendix D gives a plain-language glossary of the main technical terms.

2 Literature Review

2.1 Carbon-aware computing

Carbon-aware scheduling exploits temporal and spatial variation in grid carbon intensity. In production at Google, Radovanović et al. [20] describe a system that computes day-ahead "virtual capacity curves" to throttle flexible compute during high-carbon hours; the measured impact on latency-sensitive workloads is negligible while load shifts measurably toward cleaner hours. Wiesner et al. [23] analyse the potential of temporal shifting across regions and workloads and find the benefit depends strongly on the grid's marginal mix and the workload's flexibility. Wijayawardana and Chien [24] study cloud-VM scheduling on datacenters whose capacity varies with grid conditions, including carbon-coupled capacity, where a constant carbon budget forces capacity to fall as intensity rises, and document the goodput cost of capacity variation for long-running jobs. This systems literature establishes operational levers (deferral, throttling, capacity coupling) that we adopt as constraints, but it treats future carbon intensity as a point forecast and handles uncertainty implicitly, by re-planning. That geographic migration itself lowers carbon, moving compute toward whichever region is cleaner, is established by this line of work [20, 23], and we take the mechanism as given. What remains assumed rather than measured is whether *modelling* the joint spatial structure, or hedging forecast error robustly, adds anything on top.

2.2 Distributionally robust optimization and the price of robustness

DRO optimizes against the worst-case distribution within an ambiguity set, and the Wasserstein ball around the empirical distribution has emerged as the canonical data-driven choice, with tractable finite-dimensional reformulations [8, 18]. For costs linear in the uncertain vector, type-2 Wasserstein DRO with a Mahalanobis ground metric reduces to a mean term plus a covariance-weighted regularizer, an interpretable "mean plus $\varepsilon\sqrt{x^\top \Sigma x}$" objective solvable as a second-order cone program (SOCP) [2]. Hall et al. [9] apply Wasserstein DRO to carbon-aware data-center scheduling with virtual capacity curves, treating each region's carbon intensity as the uncertain quantity and demonstrating robustness gains over sample-average approximation. The risk functional we target, CVaR_β, is standard for tail-focused evaluation [21]. Beyond second moments, recent work robustifies the *dependence structure* itself: Fan et al. [5] place both the marginals and the *copula* within a Wasserstein ball, the natural home for the non-elliptical structure our copula schedulers target.

Robustness is never free. Bertsimas and Sim [3] framed this as the *price of robustness*: protecting against worst-case realizations trades a guaranteed degradation in nominal performance for a bounded improvement in the tail, and the operator chooses where on that frontier to sit. This lens is central to our RQ3: we do not ask whether robustification *can* help (it can, in the tail) but whether the price it charges is paid back under conditions real grids actually reach.

2.3 Modelling cross-region dependence

Cross-region dependence enters a Mahalanobis–Wasserstein model only through the off-diagonal blocks of the covariance matrix. Estimating large covariance matrices from short panels is noisy; shrinkage estimators [19] are the standard remedy and serve here as a robustness arm. Beyond second moments, dependence is fully described by the copula [11, 22]; tail-dependence coefficients quantify joint extremes that correlation cannot see, and the Gaussian copula has zero asymptotic tail dependence for any correlation below one [7], a property we use as a diagnostic benchmark. Vine (pair-copula) constructions make high-dimensional non-elliptical dependence tractable [11, 12, 14], with established sequential structure-selection procedures [13], and copula-based scenario generation for stochastic programs is well developed [15]. Whether richer non-elliptical structure (vine copulas, or copula-ambiguity DRO [5]) improves a *stochastic program* depends on whether it changes the optimal decision out-of-sample; we test this directly in Appendix B.

2.4 The gap: value has been assumed, not measured

The carbon-aware scheduling literature establishes that *temporal* shifting (deferral to clean hours) delivers measurable savings, confirmed operationally at Google and in recent studies [20, 23]. The natural extension is *spatial* shifting (migration toward clean regions), and the distributed-computing literature finds spatial shifting dominates temporal shifting in potential [23], independently motivating the transfer channel studied here. Yet the value of *modelling* that spatial structure (via cross-region covariance, copulas, or distributional robustness) has been *assumed*, not measured. The DRO literature applies sophisticated ambiguity sets to uncertain vectors without isolating the marginal value of the cross-coordinate dependence they encode, and neither literature offers a controlled mechanism for *why* spatial structure should or should not matter. Adjacent work finds simple carbon-aware heuristics capture most attainable savings, with added sophistication offering little incremental benefit [4], and the role of dependence structure in Wasserstein DRO has been studied abstractly [6]; our wedge is the controlled, pre-committed, triangulated *measurement* of the dependence layer's value. This thesis fills that gap: we build a working day-ahead scheduler that actively migrates compute and measure the savings (RQ1), isolate the value of passive modelling sophistication via a falsification test with a mechanistic explanation (RQ2), and frame robustification as a price-of-robustness trade-off, testing when that price is paid in real data (RQ3).

3 Methodology

3.1 Research design overview

The design tests whether passive (covariance-based) spatial modelling of carbon intensity adds value to robust day-ahead scheduling, and measures the value of active transfer separately. We hold the optimization machinery fixed and manipulate one object, the cross-region blocks of the covariance supplied to the scheduler, then measure the consequence on held-out tail emissions. Three primary region sets spanning the dependence spectrum (two earlier panels corroborating) serve as replications; three nested constraint regimes guard against a result driven by a particular feasible-set geometry; a robustness battery (estimators, multiple-testing correction, walk-forward) guards against statistical artifacts; and two further experiments (mean-leveling test; tail dependence) identify the mechanism. A pre-commitment discipline removes analyst degrees of freedom.

3.2 The scheduling problem

Compute is scheduled across R regions over a $T = 24$-hour day. The decision is $x \in \mathbb{R}_{\geq 0}^{R \times T}$, where $x_{r,t}$ is compute power (MW) in region r at hour t. Carbon intensity $\rho_{r,t}$ (gCO_2eq/kWh) is uncertain;

realized emissions are linear, $\langle \rho, x \rangle = \sum_{r,t} \rho_{r,t} x_{r,t}$.

3.3 Distributionally robust formulation: day-ahead forecast-error hedging

The robust layer hedges *day-ahead forecast error*. Each day the scheduler commits against a day-ahead point forecast $\bar{\rho}$ of the carbon field (the hour-of-day climatological mean on the training window, updated daily); the realized field departs by a residual $\xi = \rho - \bar{\rho}$ whose empirical distribution $\hat{\mathbb{P}}$, pooled over 2021–2024, defines the ambiguity. We minimize worst-case expected emissions over a type-2 Wasserstein ball $\mathcal{B}_\varepsilon(\hat{\mathbb{P}})$ of radius ε centred on that distribution, with Mahalanobis ground metric induced by the empirical residual covariance $\hat{\Sigma} \in \mathbb{R}^{RT \times RT}$:

$$\min_{x \in \mathcal{X}} \sup_{\mathbb{Q} \in \mathcal{B}_\varepsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{Q}}\big[\langle \rho, x \rangle\big] = \min_{x \in \mathcal{X}} \langle \bar{\rho}, x \rangle + \varepsilon \sqrt{x^\top \hat{\Sigma}\, x}, \tag{1}$$

by Wasserstein duality for linear costs [2, 8]. With the Cholesky factorization $\hat{\Sigma} = LL^\top$ the penalty becomes $\varepsilon \| L^\top \operatorname{vec}(x) \|_2$. Introducing an epigraph variable η (distinct from the hour index t), (1) is the second-order cone program

$$\min_{x \in \mathcal{X},\, \eta \ge 0} \langle \bar{\rho}, x \rangle + \varepsilon\, \eta \quad \text{s.t.} \quad \big\| L^\top \operatorname{vec}(x) \big\|_2 \le \eta, \tag{2}$$

solved to global optimality (CLARABEL). The radius ε interpolates from the deterministic LP ($\varepsilon = 0$) to increasingly conservative schedules.

Where cross-region correlation enters. Partitioning $\hat{\Sigma}$ into $T \times T$ blocks Σ_{rs} by region pair, the quadratic form splits as

$$x^\top \hat{\Sigma} x = \sum_r x_r^\top \Sigma_{rr} x_r + \sum_{r \ne s} x_r^\top \Sigma_{rs} x_s, \tag{3}$$

so the off-diagonal (spatial) blocks are the *only* channel by which cross-region dependence can alter the schedule. The experiment manipulates exactly these blocks.

3.4 A ladder of modelling layers

The thesis studies schedulers arranged by sophistication: **(0)** carbon-blind; **(1)** per-region deterministic (each region by its own mean forecast); **(2)** per-region robust (single-region Wasserstein DRO, no cross-region structure); **(3)** joint covariance (the SOCP (1) with the full spatial Σ); and **(4)** joint copula (non-elliptical dependence; Appendix B). The screening test (Section 3.7) measures the marginal value of (3) over (2). Active transfer ($\Phi > 0$, inter-region flows) is orthogonal to this passive hierarchy and is the dominant lever, studied in Sections 4.2 and 4.10.

3.5 Feasible set and constraint regimes

The feasible set $\mathcal{X}$ collects the linear constraints below; introducing an auxiliary flexible-load variable $x^{\mathrm{f}} \in \mathbb{R}_{\geq 0}^{R\times T}$, the scheduler solves (1) over

$$
\begin{aligned}
&0 \leq x_{r,t} \leq \bar{x}_{r,t}, && \forall r,t && \text{(C0: capacity)} && (4a)\\
&x_{r,t} = \alpha_r W_r\, p_{r,t} + x^{\mathrm{f}}_{r,t}, && \forall r,t && \text{(C2a: flex/inflex split)} && (4b)\\
&\textstyle\sum_t x^{\mathrm{f}}_{r,t} = (1-\alpha_r)\, W_r, && \forall r && \text{(C2b: work conservation)} && (4c)\\
&|x_{r,t} - x_{r,t-1}| \leq \Delta_r, && \forall r,t && \text{(C3: ramp)} && (4d)\\
&\textstyle\sum_{t\in\mathcal{W}} x^{\mathrm{f}}_{r,t} \geq \gamma\,(1-\alpha_r)\, W_r, && \forall r && \text{(3a: deferral deadline)} && (4e)\\
&\pi_{r,t}\, x_{r,t} \leq \bar{P}, && \forall r,t && \text{(3b: thermal/PUE)} && (4f)\\
&\textstyle\sum_{r,t} \bar{\rho}_{r,t}\, x_{r,t} \leq B && && \text{(3d: carbon budget)} && (4g)
\end{aligned}
$$

where W_r is region r's daily work (utilization fixed at 80%, $W_r = 960$ MWh); $\alpha_r \in \{0.30, 0.50, 0.75\}$ is the inflexible fraction following the fixed intraday shape $p_{r,t}$ ($\sum_t p_{r,t} = 1$); $\Delta_r = 15$ MW/h is the ramp limit; $\mathcal{W} = [0,7]$ and $\gamma = 0.20$ set the deferral deadline; and $\pi_{r,t} = \mathrm{PUE}_0 + \kappa \max(\mathcal{T}_{r,t} - \mathcal{T}_{\mathrm{set}}, 0)$ is the temperature-coupled power-usage multiplier (data, hence a constant per cell), capped at effective power $\bar{P}$. Every constraint is linear, so $\mathcal{X}$ is polyhedral and the program (1) remains an SOCP. (We follow the source labelling: C0–C3 are the base operational constraints and the **3**· series is adapted from the SoCC'25 formulation [24]; a further aggregate hourly cap $\sum_r x_{r,t} \leq p_{\max}$ is implemented but inactive in the reported regimes.)

The capacity ceiling is constant ($\bar{x}_{r,t} = 50$ MW) except under constraint **3c**, the carbon-coupled variable capacity adapted from Wijayawardana and Chien [24], which replaces it with

$$\bar{x}_{r,t} = x_{\min} + (x_{\max} - x_{\min})\, \mathrm{CFE}_{r,t}/100, \tag{5}$$

where $\mathrm{CFE}_{r,t}$ is the training-mean carbon-free-energy share, capacity falls as the grid gets dirtier. The bounds $(x_{\min}, x_{\max}) = (50, 75)$ were calibrated on training data to a loosely binding ("Goldilocks") regime: the constraint binds on 28–37% of region-hours, leaving slack and capacity margin so it shapes the schedule without freezing it.

Three nested regimes report results as a function of operational tightness: **R3** (reference) imposes C0 + C2 + C3; **R1** (lean) adds the deferral deadline 3a; and **R2** adds the thermal ceiling 3b (climate cases) or the variable-capacity ceiling 3c (interconnection cases). The carbon budget 3d is available but inactive in the reported regimes. Each regime is solved as a *complete* model with all its constraints active jointly; the nesting probes the null's robustness to operational tightness, not the effect of any single constraint in isolation.

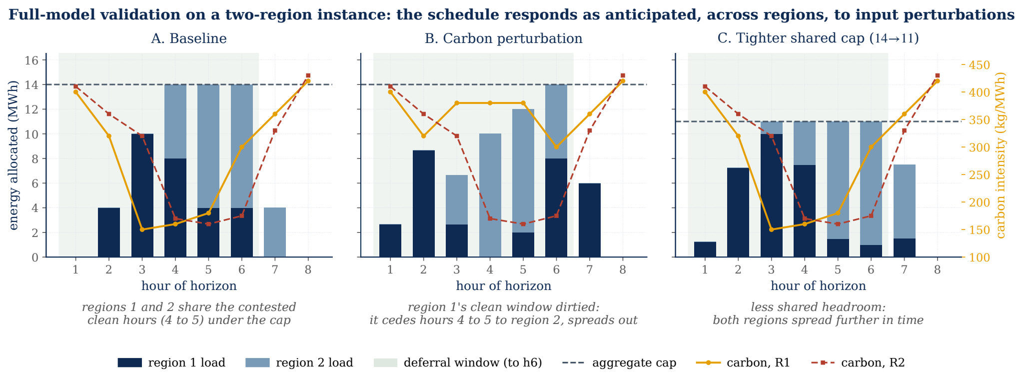

Figure 1: Full-model validation on a two-region instance. The complete feasible set (capacity, work conservation, ramp, deferral deadline) plus a shared aggregate hourly cap is solved jointly; stacked bars show how the two regions share each hour, the dashed line is the aggregate cap, the curves are each region's carbon field, the green band the deadline window. **A:** baseline, the regions split the contested clean hours (4–5) under the cap. **B:** after dirtying region 1's clean window, it cedes those hours to region 2 and spreads out. **C:** after tightening the shared cap, both regions spread further in time. Each response is the one an analyst anticipates, confirming the model encodes the intended cross-region behaviour.

3.6 Model validation on a small instance

Before running the falsification experiment we check that the optimization model behaves correctly. We validate the full model *as a whole*, with all constraints of (4) active simultaneously, on a small human-checkable *two-region* instance (eight hours, 10 MWh per-cell capacity, $W = 30$ MWh each), rather than each constraint in isolation: correctness is a property of the constraints' *interaction*, and a set that every constraint satisfies alone can still be jointly binding in unexpected ways. Two regions exercise the *cross-region* interaction, since a shared aggregate hourly cap $\sum_r x_{r,t} \leq P_{\text{agg}}$ couples the regions and forces them to compete for the same clean hours. We fix an analyst-anticipated outcome, perturb one input at a time, and verify the solved schedule moves the anticipated way (Figure 1). At baseline the two regions' clean windows overlap, and the aggregate cap forces them to split the contested hours (panel A). Making region 1's clean window dirty drives it to cede those hours to region 2 and spread to its own next-cleanest hours, with each region's work still conserved (panel B). Tightening the shared cap removes joint headroom, so both regions spread further in time rather than co-locating in the clean window (panel C). All three responses match the hand-anticipated solution and conserve work per region to machine precision. This model-level check is distinct from the software test suite (Section 3.11, which pins component behaviour such as feasible-set equivalence) and from the falsification experiment below, which probes a scientific hypothesis rather than model correctness.

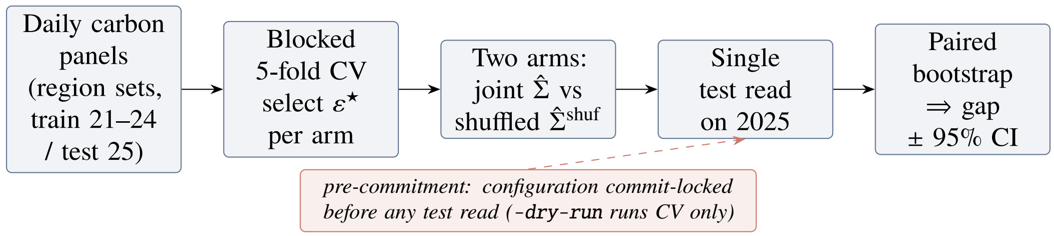

Figure 2: The pre-committed falsification pipeline. Cross-validation and schedule fitting touch only the training years; the 2025 test set is read once, after the configuration is commit-locked, and the spatial gap is read off a paired day-bootstrap.

3.7 The falsification test: shuffled marginals

The scheduler is fitted twice, with covariances differing in exactly one respect:

$$\left(\hat{\Sigma}^{\text{shuf}}\right)_{rs} = \begin{cases} \Sigma_{rr}, & r = s, \\ \mathbf{0}, & r \neq s, \end{cases} \tag{6}$$

preserving every region's marginal distribution and within-region autocorrelation while destroying only cross-region dependence. Each fitted schedule x is evaluated by the out-of-sample conditional value-at-risk of daily emissions $e_i = \langle \rho_i, x \rangle$ realized on test day i; CVaR is a coherent risk measure [16] standard in stochastic programming [17]. We use the Rockafellar–Uryasev form [21]

$$\mathrm{CVaR}_\beta(e) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \tfrac{1}{1-\beta}\, \mathbb{E}\left[(e - \tau)_+\right] \right\}, \qquad \beta = 0.95, \tag{7}$$

estimated as the empirical mean of the worst 5% of test days. The *spatial gap* is the difference between the two arms,

$$\text{gap} = \mathrm{CVaR}_{0.95}\left(e^{\text{shuf}}\right) - \mathrm{CVaR}_{0.95}\left(e^{\text{joint}}\right), \tag{8}$$

positive when the joint covariance helps. The test is a screening tool built to detect a spatial benefit if one exists; it separates the *value* of passive covariance modelling from the *validity* of cross-region correlation in the data, and checks the mean-dominance bound (Proposition 1). Figure 2 sketches the pipeline and Algorithm 1 gives the full protocol. The recipe is generic: hold the optimizer fixed, perturb exactly one modelling object, pre-commit the analysis, and read off that object's marginal out-of-sample tail value; only the perturbed object (here the cross-region covariance) changes when the same protocol is applied to another stochastic program.

Algorithm 1 Shuffled-marginals experiment (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); regimes {R3, R1, R2}; α-levels $\{0.30, 0.50, 0.75\}$; grid $\mathcal{E} = \{0, 0.1, 1, 10, 100, 1000\}$

for each regime, each α, each arm $\Sigma \in \{\text{joint}, \text{shuf}\}$ **do**

  for each $\varepsilon \in \mathcal{E}$ **do** ▷ blocked 5-fold time-series CV

    for each contiguous fold k **do**

      fit $\bar{\rho}$, L on training folds; solve SOCP (1); record validation $\mathrm{CVaR}_{0.95}$

  $\varepsilon^\star \leftarrow \arg\min_\varepsilon$ mean validation CVaR ▷ per arm, independently

  refit on full training set at $\varepsilon^\star$; evaluate *once* on 2025 ▷ single test read

paired bootstrap (1000 resamples, fixed seed) on test days $\Rightarrow$ 95% CI for the gap (8)

Pre-commitment: configuration commit-locked before any test read; `-dry-run` gate runs CV only

3.8 Data

Carbon intensity is hourly lifecycle intensity from Electricity Maps (academic licence), 2021–2025, 43,824 hourly observations per zone; panels are built on a common local clock per region set, trained on 2021–2024 and tested once on 2025. Three primary US/Canada region sets span the dependence spectrum: **(1)** the *US West*, California, Nevada and Phoenix (CISO, BANC, LDWP, NEVP, AZPS); **(2)** an *Ontario–Eastern belt* (CA-ON, NYISO, MISO, PJM); and **(3)** an *engineered heterogeneous portfolio* (CISO (solar), ERCOT (wind), BPAT (hydro), spanning *different* interconnections) deliberately chosen so clean periods do *not* align: the adversarial best case for diversification. The spread is deliberate on both ends. The two correlated, within-interconnection grids are the *best case for the spatial hypothesis*, the most cross-region correlation a covariance model could exploit, and double as a stress test rather than a realistic deployment (no operator clusters data centers within one interconnection). A realistic, globally dispersed fleet looks instead like the heterogeneous portfolio, with low and mixed correlation, so the practically relevant case is the one where diversification *should* help most. The null holding at both ends is therefore stronger than a null on either alone. A California/Nevada subset of the US West (before Phoenix was added at the supervisor's request) corroborates at the low-correlation end; an *Iberia–France* European set, explored at the project's outset, is set aside as too weak a test (its residual correlation collapses to the diurnal cycle and the DRO does not consistently engage there), so the analysis rests on the US/Canada grids. Temperature for constraint 3b is hourly ERA5 2 m air temperature (Open-Meteo), one load-weighted point per zone. The licensed raw data is never committed; derived summary tables are archived in the repository for traceability.

3.9 Validation, diagnostics, and the mean-dominance bound

Correlation validation (RQ1). Raw Pearson correlation conflates co-movement with a shared diurnal cycle. We therefore report correlations of *residuals* after removing each zone's hour-of-day mean (local clock), with overlay plots of standardized intensity for summer and winter weeks.

Robustness battery. Every experiment is re-run with (i) Ledoit–Wolf shrinkage covariance [19]; (ii) covariance estimated from seasonal (monthly-mean) residuals; (iii) covariance from AR(1) residuals. A pre-committed agreement rule counts a spatial effect only if positive and sign-stable across both residual estimators. Bootstrap p-values across all cells receive Benjamini–Hochberg correction at $q = 0.05$ [1]; a walk-forward refit (train 2021–2023, test 2024) checks temporal stability; a log-denser ε grid checks grid sensitivity.

The mean-leveling test (RQ2, mechanism). To test whether the mean field masks the covariance, the mean used for *scheduling* is transformed while evaluation stays on real emissions: LEVEL equalizes regional time-averages (keeping hour-of-day shape); FLAT sets a constant mean, making the mean term constant under fixed demand, a covariance-only world in which any joint-vs-shuf difference is attributable purely to the spatial blocks.

Tail dependence (RQ2, structural). For each region pair we estimate the upper and lower tail-dependence coefficients $\chi_{\mathrm{U}}(q) = \Pr(U > q, V > q)/(1-q)$ and $\chi_{\mathrm{L}}(p) = \Pr(U < p, V < p)/p$ on rank pseudo-observations of residual intensity, comparing χ_{U} at $q = 0.95$ with the Gaussian-copula value at the matched correlation. The Gaussian copula, the dependence family a covariance ball can represent, is asymptotically tail-independent [7], so a positive empirical excess would be exploitable tail structure invisible to the DRO; and since elliptical copulas force $\chi_{\mathrm{L}} = \chi_{\mathrm{U}}$, radial asymmetry diagnoses non-elliptical dependence.

3.10 Phase 2: richer dependence models (summary)

The covariance ball represents only *elliptical* dependence. To foreclose the "wrong object" objection, Appendix B re-runs the falsification with Gaussian, lower-tail Clayton, and the maximal (comonotone) copula, fitted to the empirical rank structure on the same feasible set. The result confirms the screening rule and is summarized in one paragraph there; Proposition 1 below explains why it must hold.

Why the dependence model is second-order. Two facts make the dependence model second-order to the mean field. The first is a decomposition. By the translation invariance of CVaR, writing the carbon field as mean plus zero-mean residual $\rho = \bar{\rho} + \xi$ and noting $\langle \bar{\rho}, x \rangle$ is deterministic,

$$\mathrm{CVaR}_\beta\big(\langle \rho, x \rangle\big) = \underbrace{\langle \bar{\rho}, x \rangle}_{\text{dependence-independent}} + \mathrm{CVaR}_\beta\big(\langle \xi, x \rangle\big), \tag{9}$$

so for a *fixed* schedule the dependence model moves only the residual term. The second is an a-priori bound on how far it can move the *optimal* value.

Proposition 1 (Spatial gap control, SOCP)**.** *Let* $\mathrm{OPT}(\Sigma)$ *be the optimal value of the SOCP* (2) *for a covariance* Σ*, and let* Σ^{shuf} *be its block-diagonal (cross-region-zeroed) counterpart with* $\Sigma^{\mathrm{off}} = \Sigma - \Sigma^{\mathrm{shuf}}$*. Then the spatial gap obeys*

$$\left|\mathrm{OPT}(\Sigma) - \mathrm{OPT}(\Sigma^{\mathrm{shuf}})\right| \;\leq\; \varepsilon \left(\max_{x \in \mathcal{X}} \|x\|_2\right) \left\|\Sigma^{\mathrm{off}}\right\|_2^{1/2}, \tag{10}$$

which vanishes as $\varepsilon \to 0$ *(no robustification) or* $\Sigma^{\mathrm{off}} \to 0$ *(no cross-region structure).*

Proof. The two programs share the feasible set $\mathcal{X}$ and differ only in the conic penalty. For fixed x, using $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a-b|}$ and the Rayleigh bound $|x^\top \Sigma^{\mathrm{off}} x| \leq \|\Sigma^{\mathrm{off}}\|_2 \|x\|_2^2$,

$$\left|\varepsilon\sqrt{x^\top \Sigma x} - \varepsilon\sqrt{x^\top \Sigma^{\mathrm{shuf}} x}\right| \leq \varepsilon\sqrt{|x^\top \Sigma^{\mathrm{off}} x|} \leq \varepsilon \|x\|_2 \|\Sigma^{\mathrm{off}}\|_2^{1/2}.$$

Since the infimum is 1-Lipschitz in the sup-norm of the objective, two objectives that differ pointwise by at most δ over a common feasible set have optimal values differing by at most δ. The feasible set $\mathcal{X}$ is a bounded polytope (the capacity bounds C0), hence compact, so the maximum $\max_{x \in \mathcal{X}} \|x\|_2$ is attained. Its value follows from combining the capacity bound $x_{r,t} \leq \bar{x}$ (C0) with work conservation (C2b): $\|x\|_2^2 = \sum_{r,t} x_{r,t}^2 \leq \bar{x} \sum_{r,t} x_{r,t} = \bar{x} \sum_r W_r = \bar{x}^2\, R\, T\, u$, so $\max_{x \in \mathcal{X}} \|x\|_2 \leq \bar{x}\sqrt{R\,T\,u}$; taking that maximum gives (10). □

The bound is a conservative *certificate*, not a tight prediction. Evaluated on each grid's training covariance (with $\varepsilon^\star = 1$, feasible bound $\max_x \|x\|_2 = \bar{x}\sqrt{R\,T\,u}$ and $\|\Sigma^{\mathrm{off}}\|_2$ the spectral norm of the destroyed cross-region blocks), its right-hand side is 7.4% (Eastern US–Canada), 9.6% (Diversified) and 12.7% (Western US) of $\mathrm{CVaR}_{0.95}$; the realized gaps (Table 1) never exceed 0.23%, more than an order of magnitude smaller (a factor of 30–55 across the three grids). The certificate is evaluated at the base per-cell cap $\bar{x} = 50$; under the variable-capacity regime 3c, where the ceiling rises to $x_{\max} = 75$, the same work-conservation argument enlarges it by only $\sqrt{x_{\max}/\bar{x}} \approx 1.22$, still well over an order of magnitude above the realized gaps. The bound holds *before any experiment is run*; what it buys is not tightness but structure, since the gap is governed by ε and the cross-region block norm and forced to zero in the no-robustification and no-cross-structure limits. We keep the a-priori form deliberately (substituting the realized $\|x^\star\|_2$ would tighten it but make it post-hoc). The *tight* ceiling is measured, not bounded: the comonotone arm of Section 4.9 realizes the supremum over copulas, and across scenario seeds its gain (the comonotone ceiling Λ) is centred on zero with $|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$; the mean-leveling test of Section 4.6 bounds the same quantity independently. Both put Λ an order of magnitude below the schedule's mean-driven value, so the mean field pins the schedule and the dependence model, elliptical or not, is immaterial.

Computational considerations. Each instance is small: the decision vector has $RT \leq 120$ entries, so the SOCP (2) solves in 0.19–0.26 s and the $S = 1000$ CVaR-SAA LP (11) in 0.9–1.3 s (CLARABEL/HiGHS, commodity laptop). The cost is in the *number* of instances: the full Phase 1 battery is on the order of 10^4 SOCP solves and Phase 2 adds 108 CVaR-SAA LPs, yet the whole study runs in under two hours on one core.

3.11 Reproducibility

All code is version-controlled (private repository, continuous-integration test suite, 202 unit tests, including a feasible-set equivalence test pinning the Phase 2 CVaR scheduler to the Phase 1 constraint set); experiment configurations are commit-locked before test-set evaluation; bootstrap and scenario seeds are fixed; and every number reported in Sections 4–5 traces to an archived summary table. Solvers: CLARABEL for the SOCP; HiGHS for the CVaR-SAA linear programs and deterministic LP baselines.

4 Results

4.1 The spatial assumption is valid (RQ1)

The premise holds. After removing each zone's hour-of-day mean, cross-region correlations of carbon intensity in the two electrically coupled US/Canada cases are strong and survive de-seasonalization: in the US West, CISO–NEVP 0.78 and CISO–LDWP 0.72 (shared solar resource and weather across the Western Interconnection); in the Ontario–Eastern belt, CA-ON–NYISO 0.73 and MISO–PJM 0.70 (weather fronts traversing the Eastern Interconnection). The engineered portfolio is the low, *heterogeneous* end rather than a zero-correlation grid: its residual pairs span 0.00 (ERCOT–BPAT), 0.29 (CISO–ERCOT) and 0.37 (CISO–BPAT), mean $\bar{r} \approx 0.2$. This heterogeneous correlation is *diversifiable* (it drives the largest mean-leveling signal, Section 4.6), unlike the US West's strong but *uniform*, common-mode correlation. Figure 3 shows the residual correlation matrices; overlay plots of standardized intensity (archived with the code) confirm visible co-movement in summer and winter weeks for the coupled cases. With the corroborating California/Nevada subset (0.65–0.78 within-state), the cases span the dependence spectrum from ≈ 0 (the engineered Diversified portfolio)

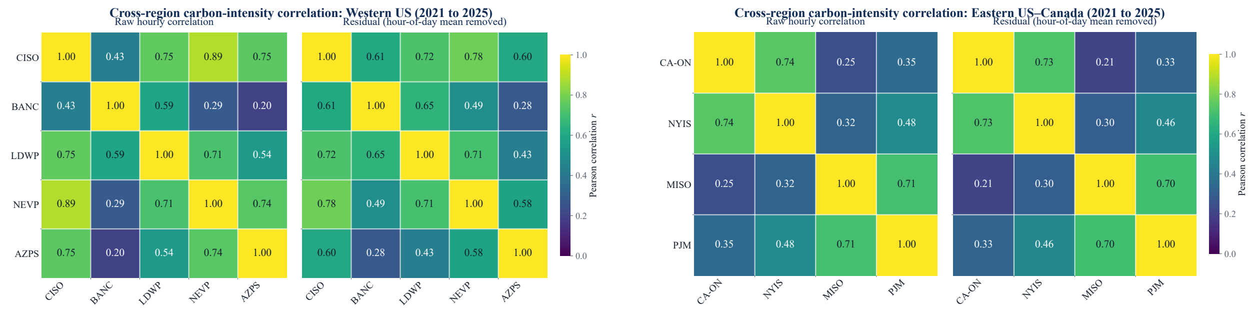

Figure 3: Residual (hour-of-day-removed) cross-region correlation of hourly carbon intensity, 2021–2025. Left: US West (CISO, BANC, LDWP, NEVP, AZPS). Right: Ontario–Eastern belt (CA-ON, NYISO, MISO, PJM). Cross-region correlation is strong and genuine in both, the assumption motivating spatial DRO is empirically valid.

to ≈ 0.8, so whatever the scheduling result, it cannot be attributed to an absence of correlation in the data.

4.2 The day-ahead scheduler and its savings (RQ1)

The scheduler migrates compute across regions against a day-ahead forecast. We report out-of-sample $\text{CVaR}_{0.95}$ on the *same* feasible set against a like-for-like comparator: carbon-aware scheduling with no inter-region transfer ($\Phi = 0$), each region scheduled by its own forecast. The transfer arm adds flows $f_{r \to s,t} \geq 0$ under a budget Φ. Unlike the Phase 1–2 falsification test, this transfer sweep and the emergency-severity crossover of Section 4.10 are exploratory extensions reported outside the pre-committed protocol.

Figure 4 reports the saving against Φ. Active transfer cuts out-of-sample $\text{CVaR}_{0.95}$ by 4.0–9.9% relative to the $\Phi = 0$ baseline across the three grids (Western 4.0%, Eastern 9.9%, Diversified 9.0%). The lever is the spatial *mean*: at each hour one region is cleanest on average, and migration ships work there. The saving is captured without any dependence model, consistent with the carbon-computing literature, which finds spatial shifting dominates temporal shifting [20, 23]. The out-of-sample reduction is largest at a moderate budget (on the US West a 4.4% peak near 10% of daily workload) and settles to the reported 4.04% plateau as the budget grows: at large budgets the deterministic schedule reallocates further toward the training-mean-cleanest region and slightly overfits out of sample, so we quote the conservative plateau as the headline rather than the peak. One caveat fixes the right reading of this number: the saving is a *gross spatial ceiling*. The transfer budget is set large enough not to bind and migration carries no per-unit, latency, or bandwidth cost, so 4.0–9.9% bounds the value of *perfect, costless* migration rather than netting out real interconnect economics; introducing a migration cost can only reduce it (limitation (i), Section 5.5).

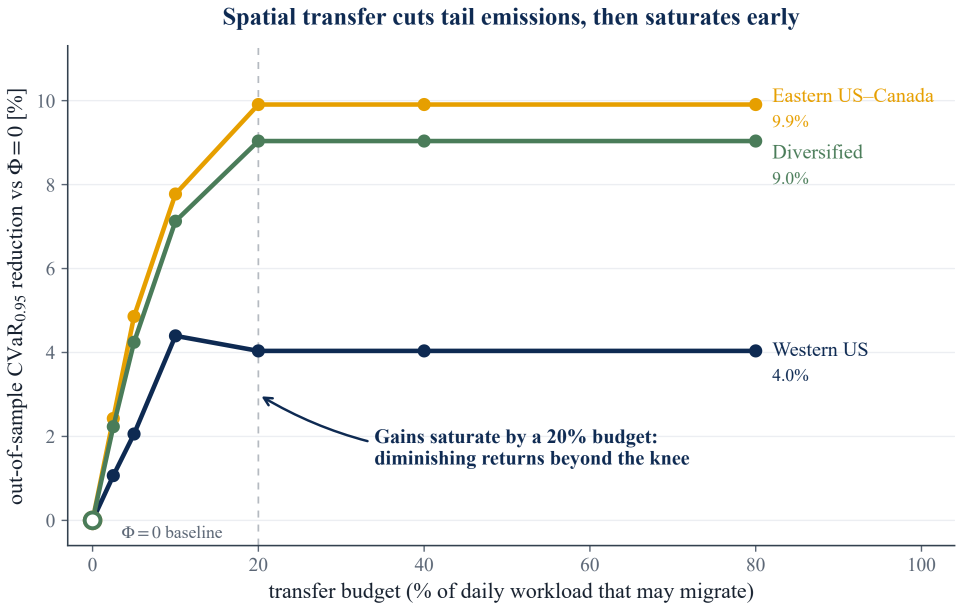

Figure 4: The transfer lever. Out-of-sample CVaR$_{0.95}$ saving against the inter-region transfer budget Φ, relative to the carbon-aware no-transfer baseline ($\Phi = 0$). Active migration captures this spatial lever; the curve saturates as the feasible mean-based gain is exhausted. The robust layer is analyzed in Section 4.10.

4.3 Where the value comes from: the complexity–value frontier (RQ1)

Figure 5 places every model class on a single complexity–value frontier (carbon-blind, deterministic per-region, robust per-region, deterministic transfer, robust transfer). Deterministic transfer dominates; the passive covariance and copula layers add no height (Sections 4.4–4.9) and robust transfer pays only in the tail (Section 4.10). Three forces, developed in Section 5.2, make the transfer saving mean-driven and keep passive covariance from adding to it: the magnitude of the diurnal mean swing, common-mode (non-diversifiable) correlation, and a decoupled dirty tail.

4.4 The screening rule: passive covariance adds nothing (RQ2)

Table 1 reports the spatial gap (8) for the three pre-committed US/Canada cases: nine cells each (three constraint regimes × three α-levels), CV-selected $\varepsilon^\star$, single test read on 2025. The gap never exceeds a few tenths of one percent of test CVaR$_{0.95}$ in either direction. Where the bootstrap calls a cell "detectable" the effect is genuine but economically negligible (on the order of a few hundred gCO_2eq/kWh out of a CVaR of 1.5×10^6) and, decisively, its *sign* flips across adjacent α-levels within the same regime: an exploitable spatial structure would not reverse direction under a change of the flexible-load fraction, so these reversals are the signature of optimization at the boundary of materiality, not of a usable effect. (We confirmed the detectable cells are not numerical: re-solving

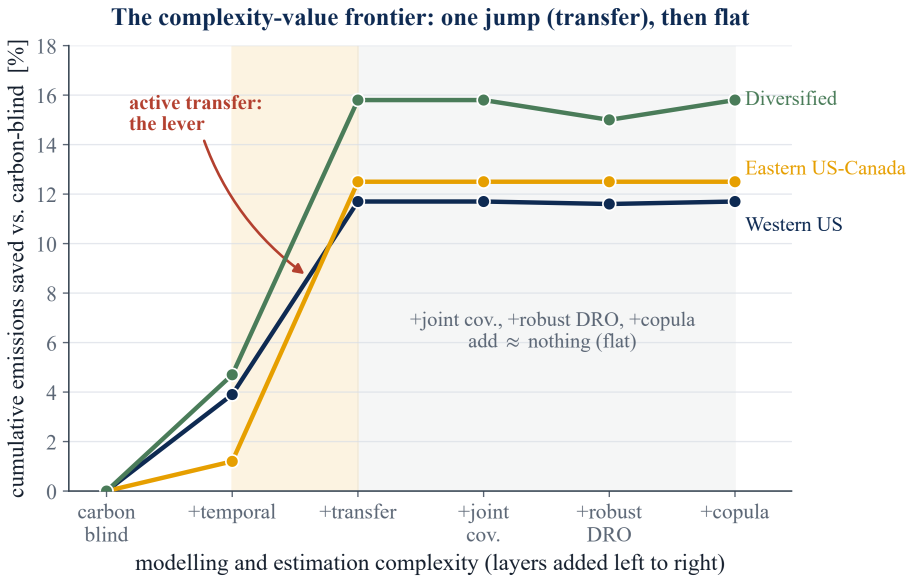

Figure 5: The complexity–value frontier. Cumulative out-of-sample emissions saved versus a carbon-blind scheduler, as each modelling layer is added. The value lives in deterministic transfer; passive dependence sophistication is flat; robust transfer is conditional on tail severity (Section 4.10). The transfer jump here is the same lever the headline states as a 4.0–9.9% reduction in tail-risk $\mathrm{CVaR}_{0.95}$ over the like-for-like $\Phi = 0$ baseline, a stricter metric than the cumulative-versus-blind axis shown.

with CLARABEL and SCS reproduces the gap to four significant figures, so the sign instability is in the data, not the solver.) The California/Nevada panel agrees (gaps in $[-0.012, +0.058]\%$, DRO active throughout).

This is a null of the *strong* kind. In all three pre-committed cases cross-validation *chooses* a non-trivial ambiguity radius: $\varepsilon^\star = 1$ in all nine cells for both Eastern US–Canada and Diversified, and in seven of nine for Western US (25 of the 27 cells overall). The DRO is therefore genuinely engaged (it is paying for robustness, not collapsing to the nominal problem) and the two arms still coincide. Had CV selected $\varepsilon^\star = 0$ the null would be uninformative, with the arms matching by construction; here the robustification is demonstrably on and the cross-region covariance still buys nothing, which is what makes the null informative rather than vacuous. Figure 6 shows this directly: validation $\mathrm{CVaR}_{0.95}$ dips to an interior minimum at a non-trivial $\varepsilon^\star$ before over-conservatism sets in, and the joint and shuffled curves coincide.

The Iberia–France panel was the project's historical starting point, and it is instructive precisely because we *set it aside*. There the residual correlation is near zero and CV does not even consistently engage the DRO: $\varepsilon^\star$ flickers between 0, 0.1, 1 and 10 across cells, collapsing to 0 in several. With no cross-region structure the machinery sees nothing worth robustifying, which makes the Iberian null the *weak* kind, a non-result rather than evidence. We therefore do not rely on it: the central

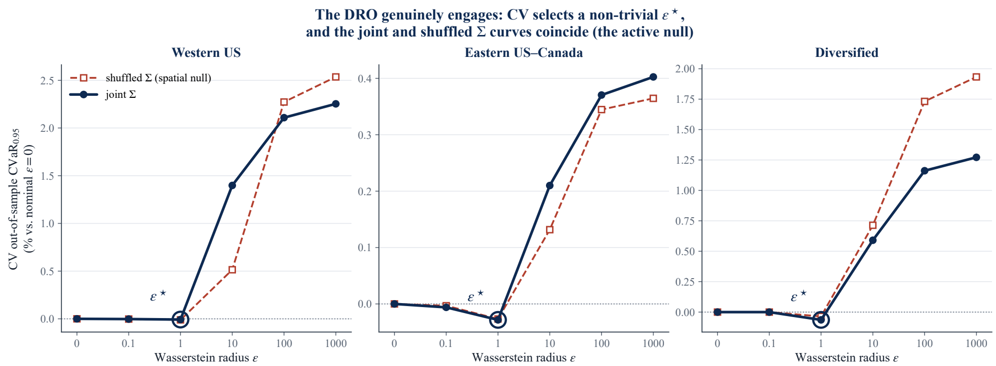

Figure 6: The DRO genuinely engages. Blocked 5-fold cross-validation $\mathrm{CVaR}_{0.95}$ (relative to the nominal $\varepsilon = 0$ problem) against the Wasserstein radius ε, for the joint (solid) and shuffled (dashed) covariances. Cross-validation selects an interior $\varepsilon^\star$ (circled), not $\varepsilon = 0$, so the robustification is on; the two curves coincide, so the cross-region covariance changes nothing. Large ε is over-conservative.

claim rests entirely on the US/Canada cases where the DRO is unambiguously engaged, and the engineered Diversified portfolio supplies the DRO-active low-correlation null ($\varepsilon^\star = 1$ throughout, still no value) that Iberia cannot.

Three readings of Table 1 matter. First, in the adversarial diversification case (Diversified) the joint covariance actually *hurts* at low α (up to -0.23%): with no true cross-structure, the joint estimator fits noise that the block-diagonal arm is spared. Second, in the strongly correlated belt (Eastern US–Canada) the detectable cells flip sign between $\alpha = 0.30$ and $\alpha = 0.75$ within the same regime. Third, the US West (strongest correlation of all) produces no detectable cell beyond one of magnitude $+3 \times 10^{-6}\%$. Figure 7A plots all 27 cells against each case's mean residual correlation: the cloud stays on zero across the whole range. **Validity of the spatial assumption does not imply value from it.**

The null is visible in the schedules themselves. Overlaying the optimal load for the joint and shuffled covariances on each region's mean carbon field, the two schedules are indistinguishable to the eye, both placing compute in the cleaner hours subject to the deadline and ramp constraints; destroying the cross-region covariance changes essentially nothing.

4.5 The null survives a pre-committed robustness battery (RQ2)

Table 3 summarizes. Re-estimating the covariance with Ledoit–Wolf shrinkage, seasonal residuals, or AR(1) residuals never produces a spatial effect that passes the pre-committed agreement rule (positive and sign-stable across both residual estimators); the largest battery value anywhere, $+0.179\%$ (AR(1), Eastern, $\alpha = 0.30$), appears only under AR(1) and is rejected. Benjamini–Hochberg correction at $q = 0.05$ across all 144 real-data cells leaves that same already-rejected

Table 1: Spatial gap (% of test $\mathrm{CVaR}_{0.95}$; positive = joint covariance better) for the three US/Canada cases. R3 = reference regime, R1 = +deadline, R2 = +variable carbon-coupled capacity. "det." = bootstrap 95% CI excludes zero.

		Diversified ($\bar{r} \approx 0.2$)		Eastern US–Canada ($\bar{r} \approx 0.45$)		Western US ($\bar{r} \approx 0.65$)	
Regime	α	gap %	det.	gap %	det.	gap %	det.
R3	0.30	−0.233	✓	−0.022	✓	+0.043	
R3	0.50	−0.177	✓	−0.002		+0.032	
R3	0.75	+0.001		+0.034	✓	−0.005	
R1	0.30	−0.233	✓	−0.022	✓	+0.014	
R1	0.50	−0.177	✓	−0.002		+0.028	
R1	0.75	+0.001		+0.034	✓	+0.002	
R2	0.30	−0.211	✓	+0.038	✓	+0.008	
R2	0.50	−0.154	✓	+0.030	✓	+0.005	
R2	0.75	+0.062	✓	+0.009		+0.000	✓

cell as the largest surviving positive gap, so nothing material survives both filters. The null also reproduces out of time (walk-forward train 2021–2023 / test 2024, maximum gap 0.217% on the Diversified grid, again negative), under an 11-point denser ε grid, and under two stress tests that strip the schedule's freedom to track the mean field, a 3× tighter ramp and a 50–95% utilization sweep: in every case the gap stays immaterial and sign-flipping with $\varepsilon^\star = 1$ retained, and the covariance does not begin to pay even when the mean's pull is mechanically curtailed. Figure 8 collects the battery: across all six arms and three grids, every spatial gap sits inside the no-value band.

Tail level: the null holds from $\mathrm{CVaR}_{0.90}$ to $\mathrm{CVaR}_{0.99}$. Because the residual co-movement lives in the *clean* tail (Section 4.7), spatial value might in principle surface only at a deeper risk level. Re-running the full pipeline at $\beta \in \{0.90, 0.95, 0.99\}$ rejects this (Table 2): the DRO stays engaged and the gap stays immaterial throughout. The largest positive gap anywhere is +0.14% (Western US, $\beta = 0.99$), a third of the margin and within one paired-bootstrap standard error of zero, while on the Diversified grid the joint covariance becomes *more* negative (−0.36%, i.e. it hurts), as the noise-fitting account predicts. This sweep is descriptive, outside the pre-committed 144-cell FDR family, and no cell is both positive and materially large.

The null is not a conditioning artifact. Block-diagonalizing Σ also changes its conditioning, so the null might reflect the joint arm being harder to estimate rather than the cross-structure being valueless. The two arms are conditioned to the same order (ridge-regularized $\kappa \approx 1.1\text{–}1.6 \times 10^5$ for the joint covariance, $0.4\text{–}0.8 \times 10^5$ for the shuffled, the joint arm somewhat higher as expected for the larger model). The mean-leveling test of Section 4.6 is the decisive control: it reuses the *same* joint Σ at the *same* conditioning and, removing only the mean field, the covariance recovers up to

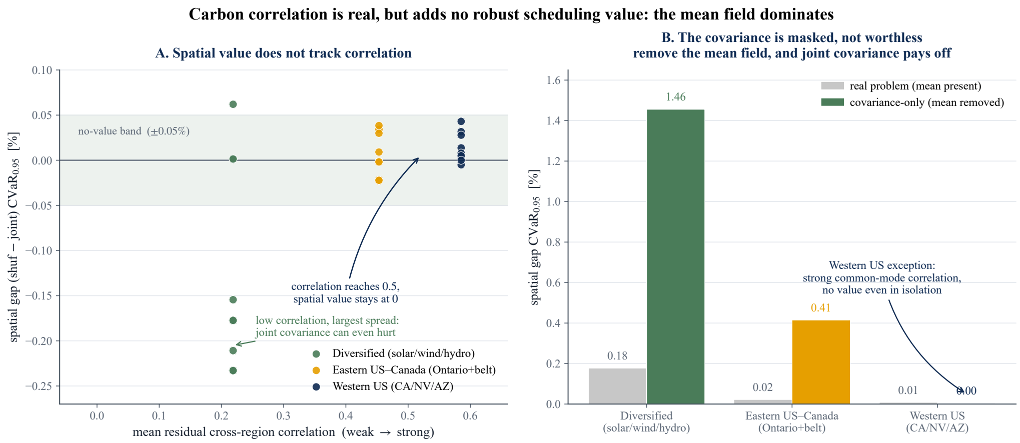

Figure 7: The Phase 1 finding. **A:** spatial gap vs. mean residual cross-region correlation; value does not track correlation anywhere on the spectrum. **B:** mean-leveling test; in the covariance-only world (constant scheduling mean) the joint covariance pays off by up to +1.46%, proving the spatial signal is real but *masked* by the mean carbon field, not absent.

Table 2: Spatial gap range (min, max over the nine cells, % of test CVaR_β) at three tail levels. The DRO is engaged ($\varepsilon^\star = 1$ almost everywhere) throughout. No level yields a sign-stable positive gap reaching the 0.4% materiality margin.

Grid	$\beta = 0.90$	$\beta = 0.95$	$\beta = 0.99$
Western US	[−0.04, +0.01]	[−0.01, +0.04]	[0.00, +0.14]
Eastern US–Canada	[−0.02, +0.04]	[−0.02, +0.04]	[−0.01, +0.09]
Diversified	[−0.23, +0.08]	[−0.23, +0.06]	[−0.36, +0.02]

+1.46%. Estimation conditioning therefore cannot be what suppresses the spatial value; the mean field is.

Statistical power: the null is not an underpowered test. The paired day-bootstrap gives a standard error on the spatial gap of SE = 0.005–0.039% of $\mathrm{CVaR}_{0.95}$ across the three grids, so the minimum benefit detectable at 80% power ($\alpha = 0.05$, two-sided) is $\mathrm{MDE}_{80} = 2.80\,\mathrm{SE} = 0.015$–0.11%, an order of magnitude below the 0.4% materiality margin and about the size of the per-cell gaps themselves. Yet no sign-stable positive gap of even this size survives: the null is a genuine absence of usable spatial value, not a test too blunt to see it.

Equivalence: the absence of a material effect is itself significant. Power bounds the effect the test *could* have missed; an equivalence test makes the affirmative claim that a material effect is *absent*. Applying the two-one-sided-tests (TOST) procedure [25, 26] per cell, against the economic materiality margin $\Delta = 0.4\%$ of $\mathrm{CVaR}_{0.95}$ (Section 5.4), all 27 cells are equivalent: every 90% interval lies strictly inside $(-\Delta, +\Delta)$. The binding cell is the adversarial Diversified case (gap

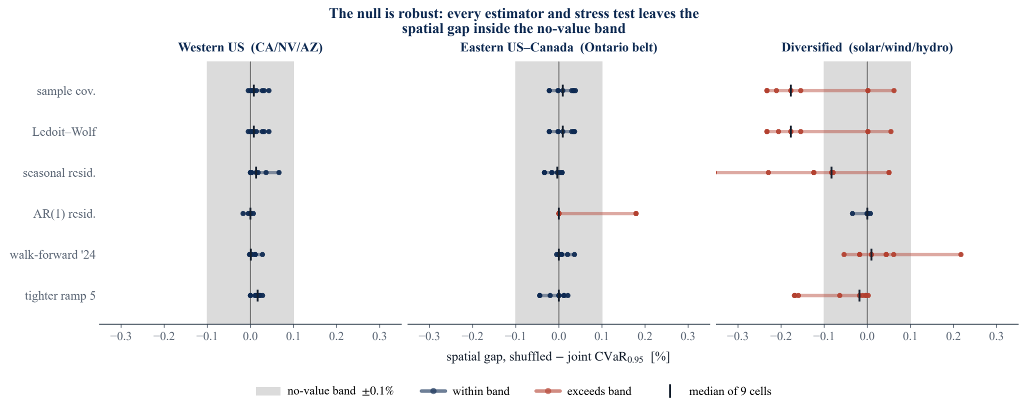

Figure 8: Robustness of the null. For each grid and each robustness arm (sample covariance, Ledoit–Wolf, seasonal and AR(1) residuals, walk-forward to 2024, and a 3× tighter ramp), dots are the spatial gap for the nine (regime, α) cells and the tick marks the median. The shaded band is the ±0.1% no-value threshold. No arm produces a sign-stable, material gap; the few Diversified excursions are negative (the joint covariance mildly hurts).

Table 3: Robustness battery: maximum |gap| (%) per covariance estimator, and temporal walk-forward (test year 2024). No cell passes the pre-committed agreement rule (positive and sign-stable under both seasonal and AR(1) residualization).

Case	Ledoit–Wolf	Seasonal	AR(1)	Walk-forward 2024
Diversified	0.233	0.355	0.035	0.217
Eastern US–Canada	0.037	0.034	0.179	0.036
Western US	0.043	0.066	0.017	0.028

= −0.233%, SE = 0.046%), which still clears both one-sided tests ($z = 3.7$, $p = 1.3 \times 10^{-4}$); every other cell gives $z \geq 7$. We therefore do not merely fail to detect a spatial benefit but statistically reject one larger than the materiality threshold, cell by cell. The margin is fixed a priori from the economic grounding, not reverse-engineered: equivalence holds for any $\Delta \geq 0.31\%$, with room to spare on the two interconnected grids and only narrowly on the engineered portfolio, where the binding gap is *negative*, so the grid where equivalence is hardest to assert is the one where covariance *hurts* rather than adds.

The null survives a serial-dependence bootstrap. Because $\mathrm{CVaR}_{0.95}$ averages the year's worst ≈12 days, which cluster under multi-day weather, an independent-day bootstrap can understate the standard error. Re-estimating the gap intervals with a moving-block bootstrap ($L = 7, 14$ days; the `block_bootstrap` snapshot) widens the standard error by 9–44% yet leaves every grid's interval well within the 0.4% margin (widest bound 0.22%), so the equivalence conclusion is not an artifact of the independence assumption.

Table 4: Mean-leveling test: spatial gap range (%) across the nine cells per case. LEVEL (equalize regional time-averages) leaves the null untouched; FLAT (constant scheduling mean = covariance-only world) reveals genuine spatial value except where correlation is common-mode (Western US).

Case	Real problem	LEVEL	FLAT (covariance-only)
Diversified	−0.233 . . . + 0.062	unchanged	+0.39 . . . + 1.46
Eastern US–Canada	−0.022 . . . + 0.038	unchanged	+0.14 . . . + 0.41
Western US	−0.005 . . . + 0.043	unchanged	−0.04 . . . + 0.00

4.6 Why the covariance is masked: the mean-leveling test (RQ2)

The mean-leveling test isolates the cause (Table 4, Figure 7B). Under the LEVEL setting (regional time-averages equalized, hour-of-day shape kept) every gap is unchanged to three decimals, so *between-region* mean differences are not the mask. Under the FLAT setting (constant scheduling mean, the covariance-only world) the joint covariance suddenly pays: up to +1.46% in Diversified and +0.41% in Eastern US–Canada, all BH-significant, and CV no longer collapses both arms onto the same schedule. The spatial signal in the covariance is therefore *real and exploitable*, but in the real problem it is dominated by the within-day shape of the mean carbon field, which both arms share. The exception is Western US: even in the covariance-only world its gaps stay at zero (−0.036 to +0.004%): its strong correlation is *common-mode*, all five zones move together, so destroying cross-dependence changes no scheduling decision. Common-mode correlation is non-diversifiable risk, with no hedge for a DRO to find.

The flat-setting result doubles as the test's *positive control*: in a world constructed so the dependence layer *should* pay, the same protocol fires (up to +1.46%), whereas on the real data it reads zero. That contrast is what licenses the main null as a true negative, a genuine absence of value, rather than a test too blunt to detect it.

4.7 Tail dependence: the structure is non-elliptical (RQ2)

Table 5 reports residual tail-dependence at $q = 0.95$ for representative pairs. Two findings. *First*, the upper tail is empirically at or below the Gaussian benchmark everywhere (excess −0.18 to +0.04): in the dirtiest hours, regions *de-couple* rather than spike together, leaving no upper-tail co-movement to exploit on the risk side. *Second*, the dependence is radially asymmetric, $\chi_L > \chi_U$: regions go *clean* together more strongly than dirty together, clearest in the solar-driven US West (CISO–LDWP $\chi_L = 0.48$ vs. $\chi_U = 0.25$; BANC–LDWP 0.40 vs. 0.17). Elliptical copulas force $\chi_L = \chi_U$, so a Mahalanobis–Wasserstein ball cannot represent this structure: for the part of the dependence that varies, the covariance is not merely masked (Section 4.6) but the *wrong object*. Figure 9 shows the asymmetry for the Ontario–Eastern belt.

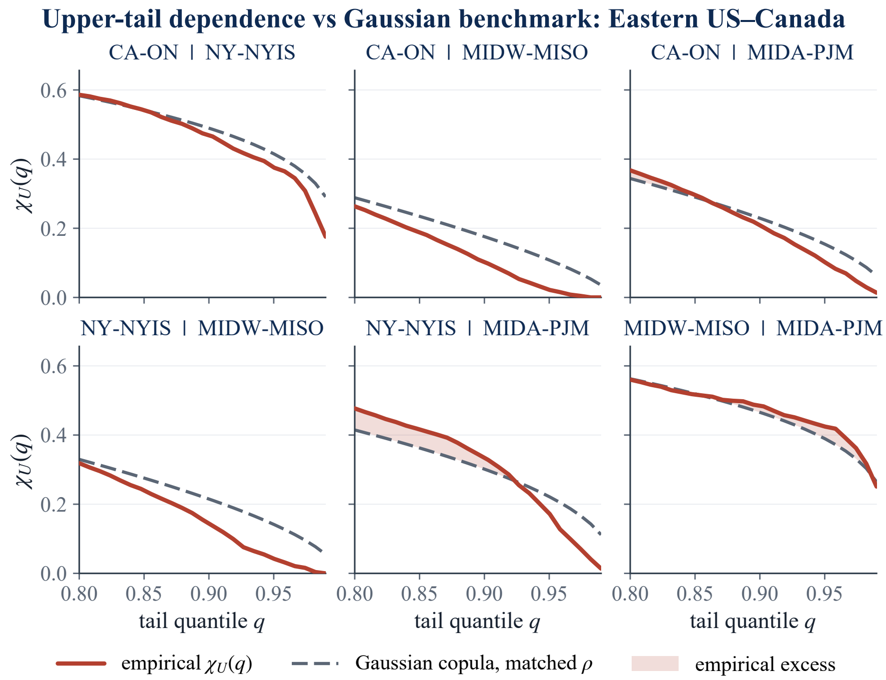

Figure 9: Upper-tail dependence $\chi_{\mathrm{U}}(q)$ for the Ontario–Eastern belt region pairs (solid) against the Gaussian-copula benchmark at the matched correlation (dashed). The empirical curves sit at or below the benchmark as $q \to 1$: in the dirtiest hours the regions decouple, so there is no exploitable upper-tail co-movement for a richer ambiguity set to capture. The complementary radial asymmetry ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$) is quantified in Table 5.

4.8 The null survives a different risk objective (RQ2)

A natural objection is that the null is a property of the particular objective we optimise rather than of carbon scheduling. The robust scheduler penalises the *standard deviation* of emissions, $\varepsilon\sqrt{x^\top \Sigma x}$, a square root that *dampens* the cross-region covariance, so perhaps an objective that weighted the covariance more heavily would let it pay. We test this directly. Holding the data, the feasible set, and the out-of-sample metric fixed, we replace the standard-deviation penalty with a raw *variance* penalty, $\lambda\, x^\top \Sigma x$ (a Markowitz-style mean-variance objective), a quadratic that *amplifies* the covariance instead of dampening it and so gives it its strongest chance to help. We then re-run the shuffled-marginals falsification, sweeping the covariance weight λ from negligible up to the point where the variance term is about 900× the mean term (`risk_measure_swap` snapshot).

Even with the covariance amplified, modelling it does not pay (Table 6). At the weight where the variance term balances the mean, the schedule shifts by up to 30% of compute relative to the no-covariance (shuffled) schedule, so the covariance genuinely drives the decision, yet using the joint covariance is no better out of sample than shuffling it away, and is in fact *adverse* on two of three grids: the joint covariance *worsens* the held-out $\mathrm{CVaR}_{0.95}$ tail by 1.5% (Western) and

Table 5: Residual tail dependence at $q = 0.95$ for selected pairs. "Excess" = empirical χ_U minus the Gaussian-copula value at the matched correlation ρ. Throughout, χ_U shows no exploitable upper-tail excess while $\chi_L > \chi_U$ signals non-elliptical (radially asymmetric) dependence.

Case	Pair	ρ	χ_U emp	χ_U Gauss	Excess	χ_L emp
Western US	CISO–LDWP	0.72	0.25	0.41	−0.16	0.48
Western US	BANC–LDWP	0.65	0.17	0.35	−0.18	0.40
Western US	CISO–NEVP	0.78	0.43	0.47	−0.05	0.39
Eastern US–Canada	CA-ON–NYISO	0.73	0.38	0.42	−0.04	0.31
Eastern US–Canada	MISO–PJM	0.70	0.43	0.39	+0.04	0.42
Diversified	CISO–BPAT	0.33	0.10	0.16	−0.06	0.26
Diversified	ERCOT–BPAT	0.00	0.04	0.05	−0.02	0.02

Table 6: Risk-measure swap. Re-running the shuffled-marginals test under a mean-variance objective (a variance penalty that amplifies covariance), at the weight where the variance term balances the mean. The schedule moves materially, but the out-of-sample $\mathrm{CVaR}_{0.95}$ gap (shuffling minus modelling joint covariance, per Eq. (8); positive means covariance helps) delivers no material benefit and is materially adverse on two of three grids. Paired bootstrap, 1000 resamples.

Grid	Schedule move	OOS $\mathrm{CVaR}_{0.95}$ gap	95% CI
Western US	∼30%	−1.50%	[−1.92, −1.18]
Eastern US–Canada	∼20%	+0.11%	[+0.04, +0.19]
Diversified	∼7%	−0.84%	[−1.02, −0.63]

0.8% (Diversified) and helps only trivially (+0.11%) on the Eastern belt. Even out-of-sample standard deviation, the quantity this objective actually targets, shows no robust covariance benefit, and in-sample variance reduction exceeds out-of-sample throughout, the classic signature of fitting estimation noise. These adverse gaps do not overturn the spatial null; they reinforce it: modelling the cross-region covariance *hurts* the held-out tail because the estimated dependence does not generalise. The null is therefore not an artifact of the robust scheduler's objective: handed an objective built to exploit covariance, the estimated dependence overfits and degrades the tail. This strengthens the screening rule rather than bounding it.

4.9 Richer dependence models confirm the screening rule (RQ2)

If the covariance ball failed only because it is elliptical, a copula that encodes the empirical $\chi_L > \chi_U$ asymmetry should recover value. It does not. Re-running the falsification with Gaussian, Clayton, and the maximal (comonotone) copula on the same feasible set leaves the screening rule intact: across scenario seeds even the maximal coupling's gain is centred on zero at the sampling-noise floor ($|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$). The binding constraint is the mean field rather than the shape of the dependence object, as Proposition 1 predicts.

Table 7: Online robust vs. deterministic transfer, rolling 2025 evaluation with no injected emergency. Out-of-sample $\mathrm{CVaR}_{0.95}$ reduction from robustifying (positive means robust is better); only the Diversified grid's interval excludes zero, and there the robust controller is significantly worse.

Grid	$\mathrm{CVaR}_{0.95}$ change	95% CI	Verdict
Western US	+0.28%	[−0.65, +1.16]	null
Eastern US–Canada	+0.04%	[−0.10, +0.25]	null
Diversified	−10.34%	[−18.3, −4.4]	robust worse

4.10 The price of robustness: when does the crossover pay? (RQ3)

Passive sophistication adds nothing under nominal conditions; the question the title poses is when *robustness* pays. We make the spatial structure an active decision (inter-region flows) and add a two-stage robust commitment with migration-limited recourse, hedging day-ahead forecast error. By Proposition 1's logic the mean dominates under nominal carbon, and indeed robust ≈ deterministic there: the DRO buys no mean reduction (a price-of-robustness result, not an emissions win), and the value of the stochastic solution is near zero at best. In the rolling online evaluation (Table 7) the robust controller's out-of-sample $\mathrm{CVaR}_{0.95}$ is statistically indistinguishable from the deterministic one on the US West (+0.28%, 95% CI [−0.65, +1.16]) and Eastern (+0.04%, [−0.10, +0.25]) grids, but is significantly *worse* on the Diversified grid (−10.34%, [−18.3, −4.4], a single representative seed whose four-seed mean is a milder but still negative −4.4%): hedging forecast error when no emergency is present costs tail performance where the regions are weakly coupled. This sharpens the decision rule: under nominal conditions robustness is at best free and can actively hurt, so it is worth buying only when emergencies past the crossover are expected.

The null holds in the deep tail. A robust controller is a tail hedge, so $\mathrm{CVaR}_{0.95}$ might understate its value for a more tail-averse operator. Re-evaluating the same online controllers at progressively deeper risk levels, from $\mathrm{CVaR}_{0.90}$ through $\mathrm{CVaR}_{0.99}$ to the single worst day (the `dro_tail_sensitivity` snapshot), the gap stays immaterial throughout: none of the twelve grid-by-metric cells shows a significant robust improvement, and on the Diversified grid robustness stays worse. The mean-dominance ceiling holds beyond the headline tail level; day-ahead robustness does not earn its keep even for a maximally tail-averse operator.

Robustness earns its keep only in the tail. Modelling rare grid-emergency events (with probability 0.1 per day a region's carbon scales by a severity M), the robust commitment's advantage over the risk-neutral one grows with M. Across four independent emergency-scenario seeds the crossover survives on the strongly correlated US West grid only: the gain is −0.4% at $M = 1$, +0.6% (all four seeds) at $M = 2$, and +4.4% of $\mathrm{CVaR}_{0.95}$ on average (s.d. 3.4%, three of four seeds) at $M = 4$ (Figure 10). On the Eastern and Diversified grids the gain hovers near zero with unstable sign and does not survive multi-seed testing, so any single-seed positive there is scenario-seed noise. The crossover is therefore real but grid-specific and only marginally significant (US West, one-sided), with $M^\star \approx 3$ the materially relevant threshold.

Real grids stay below the crossover. The synthetic multiplier overstates real risk, and a data-grounded test confirms the crossover does not activate. On the *joint* (portfolio) axis the crossover lives on, the worst realized day reaches only 1.3–1.9× the portfolio mean across the three grids (Western 1.34×, Eastern 1.29×, Diversified 1.89×; archived in `carbon_ceiling.csv`), well below $M^\star \approx 3$. Per-region *relative* swings can be larger on very clean grids (BPAT 5.1×, Ontario 2.6×), but those sit on near-zero carbon bases and do not move the portfolio tail; Winter Storm Uri, the worst US grid event on record, reached only 1.3×. Decisively, when emergencies are drawn *from real data* (each emergency replaces a region's day with a draw from its own empirical top-5% tail; `run_part3_real_emergency.py`), the robust commitment does *not* pay: its $\mathrm{CVaR}_{0.95}$ gain over the risk-neutral schedule is negative or within noise on every grid and frequency (Western −0.3%, Diversified −0.9% to −2.3%, Eastern +0.1% to +0.2%). Real emergencies are not severe enough, in joint absolute-emission terms, to reach the crossover, so deterministic transfer is dominant on observed data. We claim no universal impossibility, only that robustness does not activate across the zones and real emergencies we have.

5 Discussion

5.1 What the null means, and what it does not

The result is a *specification* finding, not a failure of carbon-aware scheduling. Carbon intensity *is* spatially correlated (Section 4.1) and the DRO *is* active ($\varepsilon^\star = 1$ almost everywhere); what the experiment falsifies is that encoding that correlation in a covariance-based ambiguity set improves robust scheduling. The mean-leveling test licenses the stronger reading: the signal is genuine (worth up to 1.46% in a covariance-only world) but subordinate to the diurnal mean field, so the null is "present but dominated", not "no signal". The central contribution is the clean separation of an assumption's *validity* from its *value*.

5.2 Why the mean dominates: a mechanism

Three forces compound. (i) *Magnitude*: the within-day swing of mean intensity (two-to-fourfold in solar-heavy grids) dwarfs the residual covariance, so the schedule is driven by *when* each region is clean on average and the second-order term moves decisions only at the margin. (ii) *Common-mode correlation*: where correlation is strongest (US West) it is positive and shared, so the risk is common

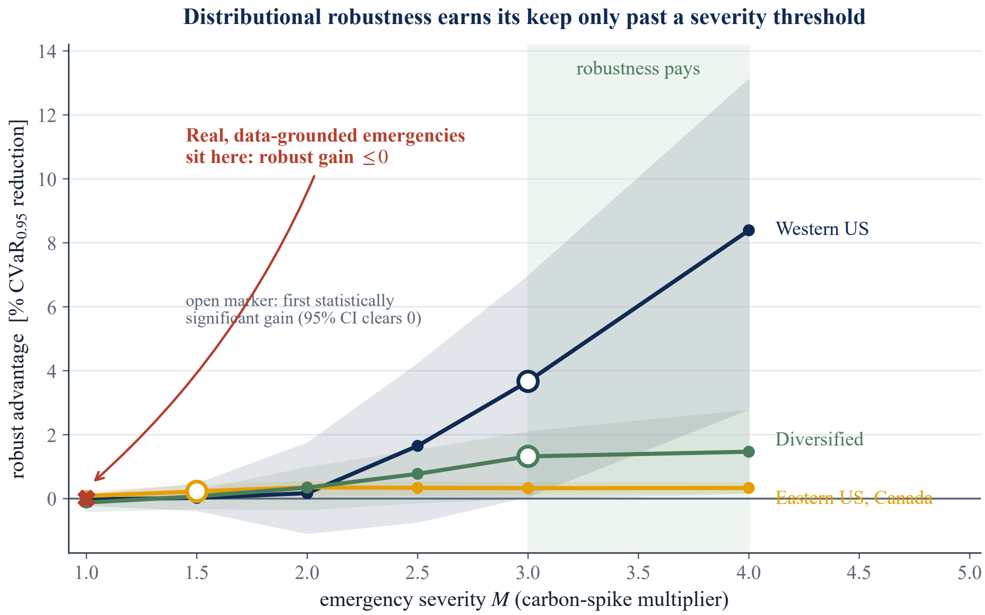

Figure 10: The price-of-robustness crossover. Robust-minus-risk-neutral $\mathrm{CVaR}_{0.95}$ advantage against synthetic single-region emergency severity M. The materially relevant crossover is $M^\star \approx 3$ (grid-specific); the plotted curves are single representative seeds, and under four independent seeds the crossover is sign-stable only on the US West grid (+4.4% at $M = 4$). Realized joint severity stays at 1.3–1.9× and the robust commitment does not pay on real top-5% emergency days, so robustness does not activate on observed data. Rule: deploy deterministic transfer; robustify only if you expect $M > M^\star$.

rather than diversifiable, and destroying the cross-blocks changes nothing. (iii) *Wrong tail*: the empirical upper (dirty) tail is close to independent while the co-movement sits in the *clean* tail ($\chi_L > \chi_U$), a regime a risk-averse scheduler need not protect and a radial asymmetry an elliptical ball cannot encode. The covariance is thus both masked (dominated by the mean) and, for its varying part, mis-shaped (non-elliptical).

5.3 Relation to prior work

The finding refines rather than contradicts the carbon-aware DRO literature. Per-region Wasserstein DRO for carbon-aware scheduling, robustifying each region's carbon intensity in isolation, is established by Hall et al. [9]; their multi-datacenter formulation does not model the joint cross-region carbon distribution. They report robustness gains over sample-average approximation; the present model is the natural *multi-region* extension that treats carbon intensity as a correlated vector. Two quantitative observations place our null in that context. First, the robustification *is* engaged: cross-validation selects a non-trivial ambiguity radius ($\varepsilon^\star = 1$, Figure 6) rather than collapsing to the nominal problem, consistent with the motivation in [9]. Second, holding that per-region (in-isolation) robustification fixed, the cross-region covariance adds nothing: the spatial gap is null throughout. In our out-of-sample 2025 test even the per-region DRO improves on the nominal (mean-greedy) schedule by at most 0.18% of $\mathrm{CVaR}_{0.95}$ across grids, itself a reflection of the mean-dominance we document, and the spatial extension adds nothing on top. The contribution is to map where the value is *not*: neither robustification depth nor spatial coupling improves on a mean-greedy per-region schedule for this problem. This spares practitioners modelling complexity that does not pay and gives theorists a precise target, namely non-elliptical tail structure that a Wasserstein ball over covariance cannot reach.

5.4 The decision rule: when does each layer pay?

The findings compose into a practitioner rule. **Layer 1 (deploy always):** day-ahead carbon-aware per-region scheduling. This is the baseline; temporal shifting captures the within-region diurnal carbon cycle. **Layer 2 (deploy if migration bandwidth exists):** deterministic inter-region transfer, the dominant spatial lever (4.0–9.9% over the $\Phi = 0$ baseline; Section 4.2). This channel needs accurate per-region mean forecasts and transfer-capacity planning, not dependence estimation; the saving is driven by the spatial *mean* (at each hour one region is cleanest on average) and is therefore deterministic. **Layer 3 (skip):** cross-region covariance or copula modelling. Both the Mahalanobis covariance ball and the richer Clayton copula add below 0.4% of $\mathrm{CVaR}_{0.95}$ and are mildly counterproductive in the diversification-friendly case (the joint estimator fits noise). The signal is genuine (mean-leveling test recovers up to +1.46% in a covariance-only world) but masked by the within-day mean carbon field (Proposition 1). **Layer 4 (conditional):** robustify the transfer against day-ahead forecast error only if grid-emergency severity realistically exceeds $M^\star \approx 3$. The robust two-stage commitment buys a small CVaR tail reduction at a mean premium (value of the stochastic solution near zero), a textbook price-of-robustness result. Realized joint severity stayed at 1.3–1.9× the portfolio mean (well below $M^\star$), and on real top-5% emergency days the robust commitment did not pay, so this layer's promised value remained unrealized.

In absolute terms, for the Eastern US–Canada facility ($\approx$ 160 MW aggregate, $\approx$ 460,000 $\mathrm{tCO_2}$/yr), the transfer lever is worth on the order of \$1.7M per year at \$80/$\mathrm{tCO_2}$, while the cross-region-covariance channel is about \$37,000 per year and not reliably positive. The value worth pursuing lives in the transfer decision and the mean forecast, not the second moment.

What the protocol buys, beyond this problem. The value-pricing protocol (fix the optimizer, perturb one modelling object, pre-commit, measure out-of-sample tail value) transfers to any correlated-uncertainty stochastic program: it lets a practitioner decide *empirically* whether a dependence, robustness, or richer-ambiguity layer earns its estimation and compute cost. Two cheap checks come first and can save the experiment outright: the mean-dominance bound (Proposition 1) flags, before any data is touched, when the mean field is large enough that no covariance layer can move the schedule materially, and the positive-control check confirms the test can detect value when value exists. Together they turn "is this layer worth modelling?" from a matter of faith into a checkable, pre-committed measurement.

5.5 Limitations

Six bound the claims. *(i) Representative workload and costless migration.* We evaluate one canonical splittable workload with stylised, uniform per-region demand, not real operator traces, and the headline transfer saving is computed *gross* of migration economics: no per-unit, latency, SLA, or network-bandwidth cost, and an unbinding transfer budget. It is therefore an upper bound on the value of migration, not a net operational figure; a positive migration cost can only reduce it. Bandwidth and latency costs for splittable batch jobs are, however, modest relative to the carbon avoided, so the conservative 4.0% plateau is likely to survive realistic migration costs, though we do not quantify this. On the modelling side, a natural worry is that the spatial null is an artifact of the chosen objective; Section 4.8 tests this directly by swapping the standard-deviation penalty for a variance penalty that favours covariance, and finds the null robust. The constraint-tightness sensitivities (Section 4.5) address the most plausible feasible-set variations: a 3× tighter ramp and a 50–95% utilization sweep both leave the null intact. A qualitatively different objective (e.g. hard per-job deadlines) remains untested. *(ii) Test horizon and non-stationarity.* The primary read is a single held-out year (2025); walk-forward to 2024, 2023, and 2022 (training only on the years before each) all reproduce the null, the largest spatial gap across the three earlier test years being 0.21% of $\mathrm{CVaR}_{0.95}$, well within the 0.4% margin, though longer multi-year rolling evaluation would strengthen external validity further. Grid carbon structure is also non-stationary: continued renewable build-out could raise cross-region correlation (shared solar) or volatility, so the "real grids stay below the crossover" reading is bounded by the period observed. *(iii) Dependence model.* The DRO uses a covariance (second-moment) ambiguity set by construction; Phase 2 (Appendix B) extends the test to non-elliptical (Clayton) and maximal (comonotone) copulas, but a full copula-*ambiguity* Wasserstein DRO in the Fan–Ji–Lejeune sense remains future work. *(iv) Geographic scope and the limit of generalization.* The region sets span the Western and Eastern Interconnections across the dependence spectrum (an Iberia–France set was explored but set aside as too weak a test), but grids with different generation mixes (e.g. hydro-thermal systems with strong seasonal storage) remain untested. The result generalizes only to grids where the diurnal mean field dominates residual cross-region covariance: mean-dominance is an empirical regularity of carbon intensity (Section 5.2), not a universal law, and a grid whose residual covariance rivalled its mean swing could behave differently. *(v) Data licence and clocks.* Carbon intensity is Electricity Maps lifecycle estimates on a common local clock per region set; alternative methodologies or clock conventions could shift point estimates, though not by the order of magnitude separating the observed gaps from materiality. *(vi) Forecast model.* The day-ahead point forecast is deliberately a transparent hour-of-day climatological mean updated daily, not a learned predictor: anchoring the robust layer to a simple, reproducible forecast keeps the price-of-robustness result a property of hedging residual error, not of any one forecaster. A learned forecaster would only shrink the residual variance the DRO hedges, and with it the already-immaterial robustness premium; quantifying that is future work.

6 Conclusions

6.1 Answers to the research questions

RQ1: how much can the scheduler save, and which lever dominates? Active inter-region migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% over the no-transfer ($\Phi = 0$) baseline across the three grids (Section 4.2). The lever is the spatial *mean*: one region is cleanest on average and migration ships work there, so the saving is deterministic and needs no dependence model.

RQ2: when does passive modelling complexity improve tail emissions? Not on any grid we tested. The out-of-sample spatial gap stays below the 0.4% materiality margin and is not sign-stable, and the null survives shrinkage, seasonal and AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward to 2022–2024, the full $\mathrm{CVaR}_{0.90}$–$\mathrm{CVaR}_{0.99}$ range, a serial-dependence block bootstrap, a per-cell equivalence test, and a swap to a covariance-amplifying objective (Sections 4.4–4.8). It is an *active* null: the DRO is engaged ($\varepsilon^\star = 1$ almost everywhere) yet covariance adds nothing. The mean-leveling test proves the signal is real (+1.46% in a covariance-only world) but masked by the mean field; the tail dependence is non-elliptical, so a covariance ball cannot see it, and a Clayton copula recovers nothing. Proposition 1 explains why: the translation invariance of CVaR makes dependence second-order to the mean. The screening rule: passive sophistication cannot pay where the diurnal mean field dominates cross-region dependence.

RQ3: when does robustifying pay, and do real grids reach that regime? The robust two-stage commitment buys a small tail reduction at a mean premium (value of the stochastic solution near zero) and materially beats the risk-neutral schedule only past an emergency-severity crossover $M^\star \approx 3$ (Section 4.10). Realized joint severity reached only 1.3–1.9× the portfolio mean, and on real top-5% emergency days the robust commitment did not pay, so the layer's value is characterized but unrealized on observed data. We claim no universal impossibility, only the tested one; robustness remains a conditional option given a concrete emergency model.

6.2 Contributions

In summary, the thesis contributes: (1) a working day-ahead migration scheduler saving 4.0–9.9% over the $\Phi = 0$ baseline with a layered decision rule; (2) a pre-committed, falsification-based methodology separating the *validity* of a spatial modelling assumption from its *value*, transferable to any correlated-uncertainty DRO; (3) a replicated, robustness-checked null on whether spatial dependence improves carbon-aware DRO scheduling, across real grids and the covariance-to-copula hierarchy; (4) a mechanistic explanation (the mean-dominance bound, demonstrated by the mean-leveling test) that supplies the screening rule; and (5) a reproducible, CI-tested pipeline (202 tests) with archived summary tables for every reported number.

6.3 Future work

Phase 2 settled the immediate question, whether a copula that sees the asymmetry recovers value (it does not), and sharpened the remaining agenda. A *copula-ambiguity Wasserstein DRO* in the Fan–Ji–Lejeune sense [5], built on a fitted vine [12, 13], would robustify the dependence structure itself rather than sample from one fitted copula; Proposition 1 predicts the mean field will still dominate, but the claim would then be fully model-free. The body establishes the transfer savings and the severity crossover (Sections 4.2, 4.10); the open problem is to robustify the recourse policy with affine decision rules and calibrated emergency models. We are not aware of prior work that combines Wasserstein-DRO robustification with active inter-region transfer, and making that combination pay under realistic forecast error is an open question. The transfer model also invites an *online, multistage* treatment with forecasts arriving sequentially and a Wasserstein ball centred on a learned forecast, and the mean-dominance bound generalizes into a problem-class condition for *when* cross-region dependence can improve a robust allocation. The null is therefore not a dead end but a sharper target: value, if any, comes from an active spatial decision rather than a better passive dependence model.

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A Reproducibility Details

Scenario-count stability and the comonotone noise floor. Figure 11 re-evaluates the comonotone (upper-Fréchet) gap on Eastern US–Canada over five scenario seeds and a range of scenario counts S. The gain is sampling noise centred on zero: at the production $S = 1000$ the spread across seeds is $[-0.08, +0.10]\%$ with mean $+0.01\%$, and it narrows with S. This is the basis for treating the single-seed $+0.182\%$ comonotone gap of Section 4.9 as immaterial rather than a real effect, and it confirms $S = 1000$ is sufficient.

Repository and environment. All code lives in a private, version-controlled repository with a continuous-integration suite (202 unit tests in total; CI runs 189, excluding the 13 Electricity-Maps integration tests that require licensed data). Experiments run under Python with CVXPY; solvers are CLARABEL for the second-order cone program and HiGHS for deterministic LP baselines. Exact dependency versions are pinned in a tracked `requirements-lock.txt` (CLARABEL 0.11.1,

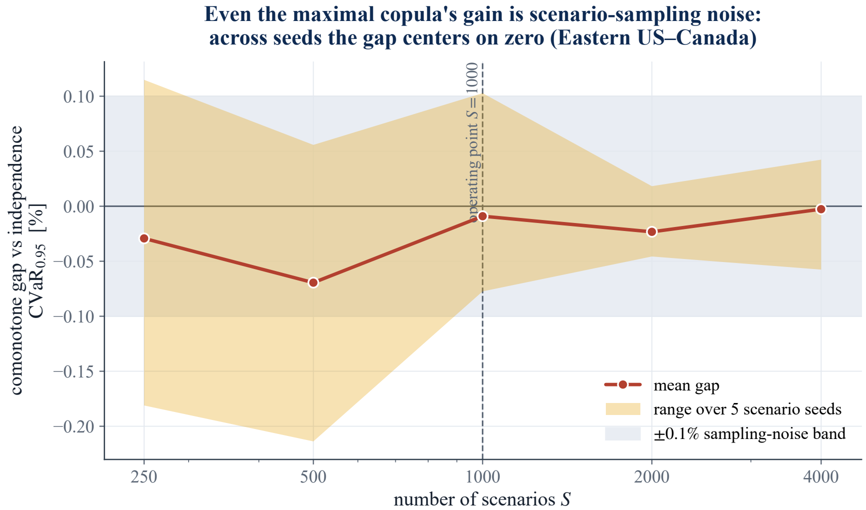

Figure 11: The comonotone gap is scenario-sampling noise. Mean (line) and range over five seeds (band) of the comonotone-vs-independence $\mathrm{CVaR}_{0.95}$ gap on Eastern US–Canada, against the scenario count S. The band straddles zero and narrows with S; the ±0.1% no-value band is shaded.

HiGHS/highspy 1.14.0, CVXPY 1.9.1, NumPy 2.4.6, SciPy 1.17.1; Python 3.12.4), under which every reported number reproduces byte-identically. The licensed raw carbon-intensity data is never committed; only derived aggregate summary tables (CV-selected ε, test CVaR, spatial gaps, bootstrap CIs, BH p-values) are archived, so the non-redistribution academic licence is respected while every reported number remains traceable.

Pre-commitment protocol. Each experiment follows commit-lock → dry-run → single test read. The full configuration (utilization 0.80; $\alpha \in \{0.30, 0.50, 0.75\}$; ε grid $\{0, 0.1, 1, 10, 100, 1000\}$; 5-fold blocked time-series CV; 1000 bootstrap resamples at a fixed seed; CVaR level 0.95; train 2021–2024, test 2025) is committed before any test-set evaluation, and a `-dry-run` gate runs cross-validation only. The 3c capacity bounds $(x_{\min}, x_{\max}) = (50, 75)$ were Goldilocks-calibrated on training data alone (bind fraction 0.28–0.37).

Reproducing the experiments. Every result regenerates with

```
python -m scripts.run_case_experiment -region-set <case> [flags]
```

where `<case>` is one of the internal region-set keys `us_west` (Western US), `taskc` (Eastern US–Canada), or `us_hetero` (Diversified), and flags select the estimator (`-shrinkage`, `-residualize`), mean-leveling test (`-ablate-mean`), walk-forward (`-test-year`), or grid density (`-eps-grid`). Snapshot files are named `<case>_regimes_<date>[_<variant>].csv`, and the snapshot `README` maps each to its source command. Table 8 maps each results subsection to its archived table.

Table 8: Mapping of reported results to archived summary tables.

Section	Result	Archived source
4.4	Spatial gaps (Table 1)	{case}_regimes_2026-06-10.csv
4.5	Robustness (Table 3)	{case}_..._{lw,seasonal,ar1,ty2022,ty2023,ty2024}.csv
4.5	BH correction	bh_correction.csv
4.6	Mean-leveling test (Table 4)	{case}_..._ablate-{level,flat}.csv
4.8	Risk-measure swap (Table 6)	risk_measure_swap_2026-06-27.csv
4.5	Serial-dependence (block) bootstrap	block_bootstrap_2026-06-27.csv
4.1,4.7	CA/Iberia panels	taskA_.. es_pt_fr_..csv
4.2	Transfer value, curve	transfer_value_curve_.., part3_transfer_value_..
4.10	Crossover, realized severity	part3_{emergency,real_emergency}_.., carbon_ceiling_..
4.10	Online robust (Table 7)	part4_online_.., dro_tail_sensitivity_..

B Richer dependence models: confirming the screening rule

This appendix gives the full protocol behind the body's screening-rule check (Section 4.9). The covariance test answers whether *elliptical* dependence helps; because the tail diagnostic flags non-elliptical structure ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$, Table 5), a natural objection to the Phase 1 null is "you used the wrong dependence object." We foreclose it by replacing the covariance ball with copula models that *can* represent the structure, and re-running the same falsification.

Copula-coupled resampling. We generalize the shuffled-marginals test from second moments to the full copula while preserving every region's empirical marginal. Each region r keeps its pool of whole daily profiles $\{\rho_r^d \in \mathbb{R}^T\}$; a scenario is generated by drawing a copula sample $u \in [0,1]^R$ and, for each region, selecting the training day whose daily-mean intensity has rank u_r (low rank = clean day), then taking that day's full profile. The copula thus governs *only* the cross-region coupling, exactly as the shuffled covariance did. Three nested fitted models span the dependence hierarchy, and a fourth, the comonotone upper bound, is added as the ceiling: **(i)** the *independence* copula (regions decoupled, the Phase 1 shuffled arm); **(ii)** the *Gaussian* copula at the empirical rank-correlation (elliptical, what the covariance ball can see); **(iii)** an exchangeable *Clayton* copula, with lower-tail dependence $\lambda_L = 2^{-1/\theta}$ and zero upper-tail dependence [11], fitted by matching mean pairwise Kendall's τ ($\theta = 2\tau/(1-\tau)$), the object built for the $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ finding; and **(iv)** the *comonotone* (upper-Fréchet) copula, the maximal coupling. Clayton is sampled by Marshall–Olkin frailty; no parametric marginal model is introduced. The exchangeable Clayton imposes a single θ and so cannot reproduce the pair-level heterogeneity of the tail diagnostic; a regular-vine copula with edge-specific families would. We do not fit one, for a principled reason: the comonotone arm realizes the supremum over *all* copulas, vines included, so it upper-bounds whatever any heterogeneous-dependence model could achieve. If the maximal-coupling arm recovers no material value, no vine can.

Copula CVaR scheduler. For each model we draw $S = 1000$ scenarios $\{\rho^s\}$ and solve the sample-average CVaR program in Rockafellar–Uryasev form [21],

$$\min_{x \in \mathcal{X},\, \tau,\, z \geq 0} \ \tau + \frac{1}{(1-\beta)S} \sum_{s=1}^{S} z_s \quad \text{s.t.} \quad z_s \geq \langle \rho^s, x \rangle - \tau, \tag{11}$$

over the *same* feasible set $\mathcal{X}$ (4) and regimes as Phase 1, so the only thing that varies across arms is the dependence model. Each schedule is evaluated once on the 2025 panel by out-of-sample $\mathrm{CVaR}_{0.95}$; the reported copula gaps are $\mathrm{CVaR}(\text{indep}) - \mathrm{CVaR}(\text{Gaussian})$, $\mathrm{CVaR}(\text{indep}) - \mathrm{CVaR}(\text{Clayton})$, and the asymmetry increment $\mathrm{CVaR}(\text{Gaussian}) - \mathrm{CVaR}(\text{Clayton})$, each with a paired day-bootstrap CI. Algorithm 2 summarizes the procedure.

Algorithm 2 Copula-scenario CVaR scheduling (one region set)

```
Input: daily panels ρ_{1:N} (train 2021–2024, test 2025); dependence models {independence,
  Gaussian, Clayton, comonotone}; regimes; α-levels; scenarios S = 1000, fixed seed
for each dependence model C do
    fit C on training daily summaries (rank pseudo-observations); for Clayton, θ = 2τ/(1 − τ)
  from mean Kendall τ
    for each regime, each α do
        draw S copula samples u^s ∈ [0, 1]^R        ▹ indep./Gaussian/Clayton frailty/comonotone
        map each region's u^s_r to the training day of that daily-summary rank; take its full profile
  ⇒ scenario ρ^s ∈ ℝ^{R×T}
        solve the CVaR-SAA program (11) over 𝒳 ⇒ schedule x
        evaluate once on 2025 ⇒ out-of-sample CVaR_{0.95}
copula gaps vs. independence + paired day-bootstrap CIs; comonotone realizes sup_C, the ceiling
  Λ
```

Result. Table 9 reports the copula gaps. Across the three grids no dependence model recovers material value: the Gaussian and Clayton gaps over independence stay below 0.2% of $\mathrm{CVaR}_{0.95}$, and the asymmetry increment (Clayton over Gaussian, the object built for the $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ structure) is smaller still. The comonotone ceiling Λ is scenario-sampling noise centred on zero ($|\Lambda| \lesssim 0.1\%$; Appendix A, Figure 11). The screening rule survives the richest passive dependence model available, exactly as Proposition 1 predicts: the binding constraint is the mean field, not the shape of the dependence object.

Table 9: Copula gaps (% of test $\mathrm{CVaR}_{0.95}$): out-of-sample improvement of each dependence model over the independence (shuffled) arm, median over the nine (regime, α) cells. No model reaches the 0.4% materiality margin.

Case	Gaussian	Clayton	Clayton−Gaussian
Western US	0.066	0.038	0.089
Eastern US–Canada	0.158	0.014	0.182
Diversified	0.069	0.127	0.114

C Extended outlook: online and multistage robust transfer

The day-ahead migration scheduler, its savings, and the severity crossover are reported in the graded body (Sections 4.2, 4.10). This appendix records implementation and validation detail for the transfer module and sketches the planned online/multistage extensions that lie beyond the capstone proper.

This appendix sketches where the line goes next, with *preliminary, proof-of-concept* experiments. They are prototype-grade, a stylised grid-emergency stress model, modest scenario counts, no pre-commitment, and a rigorous treatment is the subject of a follow-on study; we include them only because they already complete the arc the capstone opened. The capstone showed that *passive* spatial modelling adds no value because the mean dominates. The natural question is whether making the spatial structure an *active decision* changes that. We add inter-region load flows $f_{r\to s,t} \geq 0$ so the executed load $y_{r,t} = x_{r,t} + \sum_s (f_{s\to r,t} - f_{r\to s,t})$ can migrate toward the cleaner region; the program stays an SOCP/LP.

Three findings emerge (Figure 12). **(1) Active transfer unlocks spatial value.** A transfer-budget sweep cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% across the three grids, the value the passive null left on the table, captured by exploiting the spatial *mean* (shipping work to the region cleanest on average). **(2) Robustness is worthless under normal carbon, even here.** A one-shot transfer DRO, a forecast-error (persistence) ambiguity set, and a two-stage robust commitment with costly recourse all leave robust ≈ deterministic, the mean-dominance of the main text extends into the active setting. **(3) But robustness pays under tail risk.** Modelling rare grid-emergency events (with probability 0.1 per day a region's carbon is scaled by a severity M, as in a renewable drought or peaker dispatch) and re-solving the two-stage problem with migration-limited recourse traces a sharp *crossover*: at $M = 1$ the robust commitment is no better than the risk-neutral one (the main null), but as M grows it increasingly wins, by up to +4.4% of $\mathrm{CVaR}_{0.95}$ (four-seed mean, s.d. 3.4%) at $M = 4$ on the US West, the only grid where the crossover survives multi-seed testing. The robust commitment stays hedged while the risk-neutral one is burned when its favoured region spikes.

The synthesis is a clean phase diagram: distributional robustness is immaterial for carbon-aware

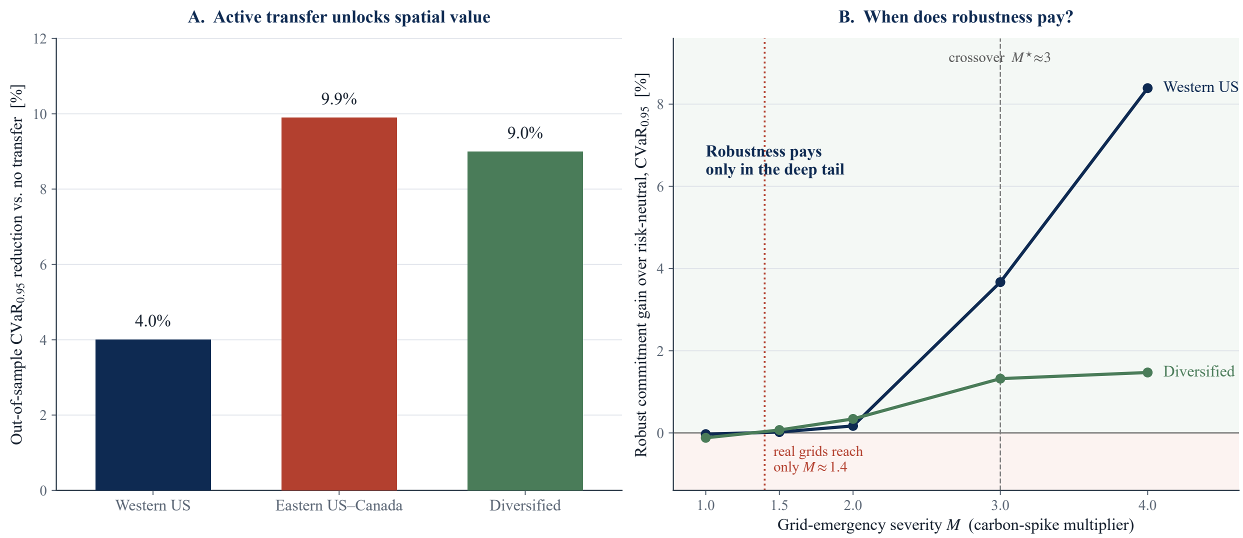

Figure 12: Transfer module validation. **A:** active transfer reduces out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% over the $\Phi = 0$ baseline. **B:** the severity crossover, the robust (CVaR-hedged) commitment's advantage over the risk-neutral one as grid-emergency severity M grows; it turns materially positive only past $M^{\star} \approx 3$. The same numbers are reported and interpreted in the body (Section 4.10).

scheduling under normal conditions (the mean dominates, whether the spatial structure is modelled passively or acted on actively), but becomes valuable precisely when rare tail events make the mean forecast unreliable. The value of robustness is thus a function of tail risk, the precise characterisation of which, together with a calibrated emergency model and a pre-committed evaluation, is the agenda for the follow-on work.

Implementation and validation. These results are not throwaway script output: the transfer model is implemented as a standalone, unit-tested module (`src/models/transfer_dro.py`), where the one-shot transfer DRO, the two-stage robust commitment, and the costly-recourse evaluation are each exercised by tests pinning the model's invariants, conservation of work under inter-region flows ($\sum_{r,t} y_{r,t} = \sum_r W_r$), monotone emission reduction as the transfer budget grows, and exact recovery of the no-transfer schedule at zero budget. The flow variables $f_{r\to s,t}$ carry no self-loops and respect a transfer-capacity budget, and the two-stage program is solved as an LP per scenario with a shared CVaR epigraph. What remains *preliminary* is therefore the experimental protocol, the stylised emergency model, the modest scenario counts, and the absence of pre-commitment, not the optimization itself: the crossover of Figure 12 rests on validated code. This is what makes promoting Part 3 to a rigorous study a matter of protocol and calibration, the steps below, rather than of rebuilding the machinery.

Directions beyond this capstone. Two extensions follow naturally and are left as future work. First, taking the transfer model *online*: a rolling-horizon robust controller in which forecasts arrive sequentially and demand and transmission are themselves uncertain, with a Wasserstein ball centred on a learned forecast and evaluated closed-loop on real operational traces. Second, *generalising* the mean-dominance bound into a problem-class condition for when cross-coordinate dependence can improve a robust allocation, of which carbon-aware scheduling is one instance among others (energy storage, EV fleets, spatial supply chains). Promoting the present Part 3 sketch to a rigorous, pre-committed study with a calibrated emergency model is the prerequisite for both. These directions are recorded in the project's research roadmap.

D Glossary: plain-language terminology

The main technical terms used in this thesis, glossed in plain language; the precise definitions are in Section 3.

Carbon intensity: how dirty the electricity is at a given hour and place (grams of CO_2 per kWh); it swings with how much wind, solar, and hydro are on the grid.

Distributionally robust optimization (DRO): planning against the worst-case carbon distribution within a set of plausible distributions around the data, rather than trusting a single point forecast. Like packing for weather a little worse than the forecast.

Wasserstein distance / ball: a way to measure how far apart two probability distributions are (the cheapest cost of reshaping one into the other). The Wasserstein *ball* of radius ε is the set of all distributions within that distance of the data, every future close enough to what we observed to be worth hedging.

$\mathrm{CVaR}_{0.95}$ **(Conditional Value-at-Risk):** the average emissions on the worst 5% of days; a tail-risk measure, not the average day.

Second-order cone program (SOCP): the convex problem the scheduler reduces to, solved to global optimality in a fraction of a second.

Mahalanobis distance: a distance that weights directions by the carbon covariance, so it accounts for how regions move together. Like measuring distance on a stretched map where correlated regions sit closer.

Copula: the part of a joint distribution that captures only how regions move together, with each region's own behaviour stripped out.

Comonotone: the strongest possible copula, all regions rising and falling in perfect lockstep; used as a worst-case upper bound.

Tail dependence (χ_U, χ_L): whether regions hit their dirtiest (χ_U) or cleanest (χ_L) hours at the same time, a joint-extreme effect plain correlation can miss.

Residualization: removing each region's predictable hour-of-day (or day-to-day) average so the analysis works only on the genuine surprises, not the routine daily cycle.

Shuffled-marginals test: re-fitting the scheduler with the cross-region link destroyed but each region's own pattern intact, so the difference isolates the value of cross-region correlation.

Mean-leveling test: we deliberately flatten each region's average carbon level (so the only signal left is the cross-region co-movement) and re-run the scheduler. It shows whether that cross-region structure is genuinely useful but normally hidden behind the much larger differences in average carbon.

Mean-dominance (Proposition 1): an a-priori bound showing the average carbon field caps how much any dependence model can change the schedule.

Out-of-sample / walk-forward: scoring a plan only on data it never trained on (out-of-sample); walk-forward is the rolling version, train on earlier years and test on a later untouched one, mimicking real deployment.

Equivalence test (TOST): a test that affirmatively shows an effect is smaller than a chosen materiality margin, turning "we found nothing" into "any effect this large is ruled out."

Severity M: the multiple by which one region's carbon spikes in a rare grid emergency ($M = 3$ is a tripling); the crossover $M^\star$ is where robustness starts to pay.

Value of the stochastic solution (VSS): how much planning against the full uncertainty distribution beats committing to a single point forecast; here it is near zero.

Transfer budget (Φ): how much compute may migrate between regions; $\Phi = 0$ is the honest no-transfer baseline, and raising it delivers the headline savings.

Mean-variance objective: penalising the spread (variance) of emissions directly, Markowitz-style, instead of the worst-case tail; used here only as a stress test that deliberately gives the cross-region covariance its best chance to help.

Positive control: a deliberately built case where the effect being tested truly exists (here, the mean-leveling world), used to confirm the test can detect value when value is present, so a null elsewhere is a real absence and not a blind test.

Translation invariance: a property of CVaR whereby adding a fixed amount to every outcome raises the risk number by exactly that amount, so a known average carbon level cancels out; this is why the mean field, not the dependence, drives the schedule.

Annex A: Individual Contribution Statement

This Research Capstone was completed individually by **Marco Ortiz Togashi** under the supervision of Prof. Bissan Ghaddar. As sole author, I am responsible for all components of the work:

- **Research design.** Formulated the three research questions, designed the shuffled-marginals falsification test, and specified the pre-commitment protocol that separates assumption validity from value.
- **Modelling.** Specified the Mahalanobis–Wasserstein DRO scheduling model and its SOCP reformulation, the constraint regimes (including the variable carbon-coupled capacity constraint adapted from the SoCC'25 formulation), and the mean-leveling test and tail-dependence diagnostics.
- **Data and implementation.** Assembled the multi-grid carbon-intensity and temperature panels, implemented the full experimental pipeline, the robustness battery, the Benjamini–Hochberg correction, and the figures, under version control with a continuous-integration test suite.
- **Analysis and writing.** Produced and interpreted all results, established the mechanistic explanation, and wrote the manuscript in full.

Generative AI tools were used as a coding and drafting assistant under my direction, as declared on the cover page; all research questions, modelling decisions, experimental designs, interpretations, and final text are my own. The inter-region migration mechanism itself follows established carbon-aware practice (Section 2); my individual contribution, developed under the supervisor's direction, is its distributionally robust treatment, the honest $\Phi = 0$-anchored measurement of its spatial value, and the screening and price-of-robustness decision rules.

Marco Ortiz Togashi   Date